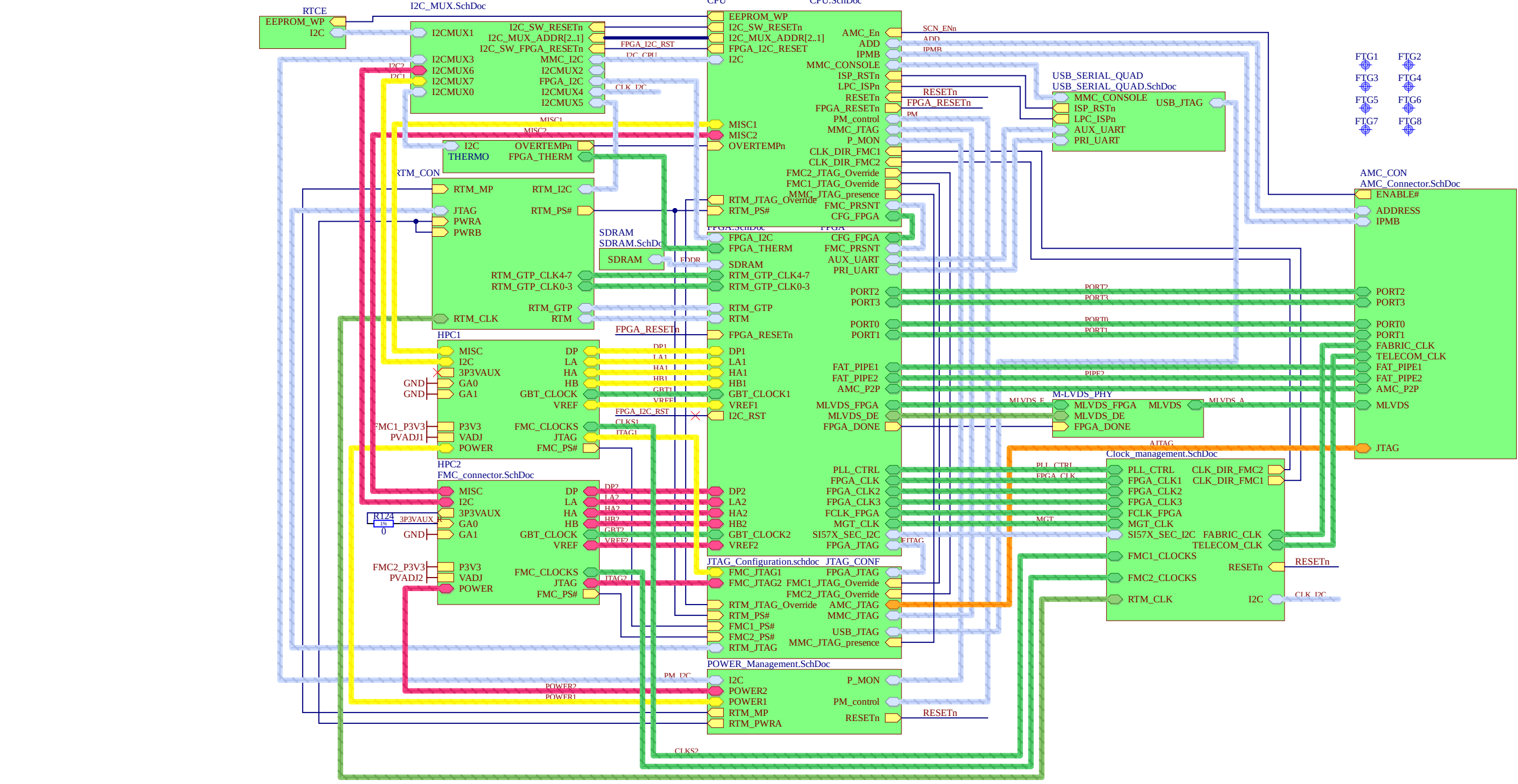




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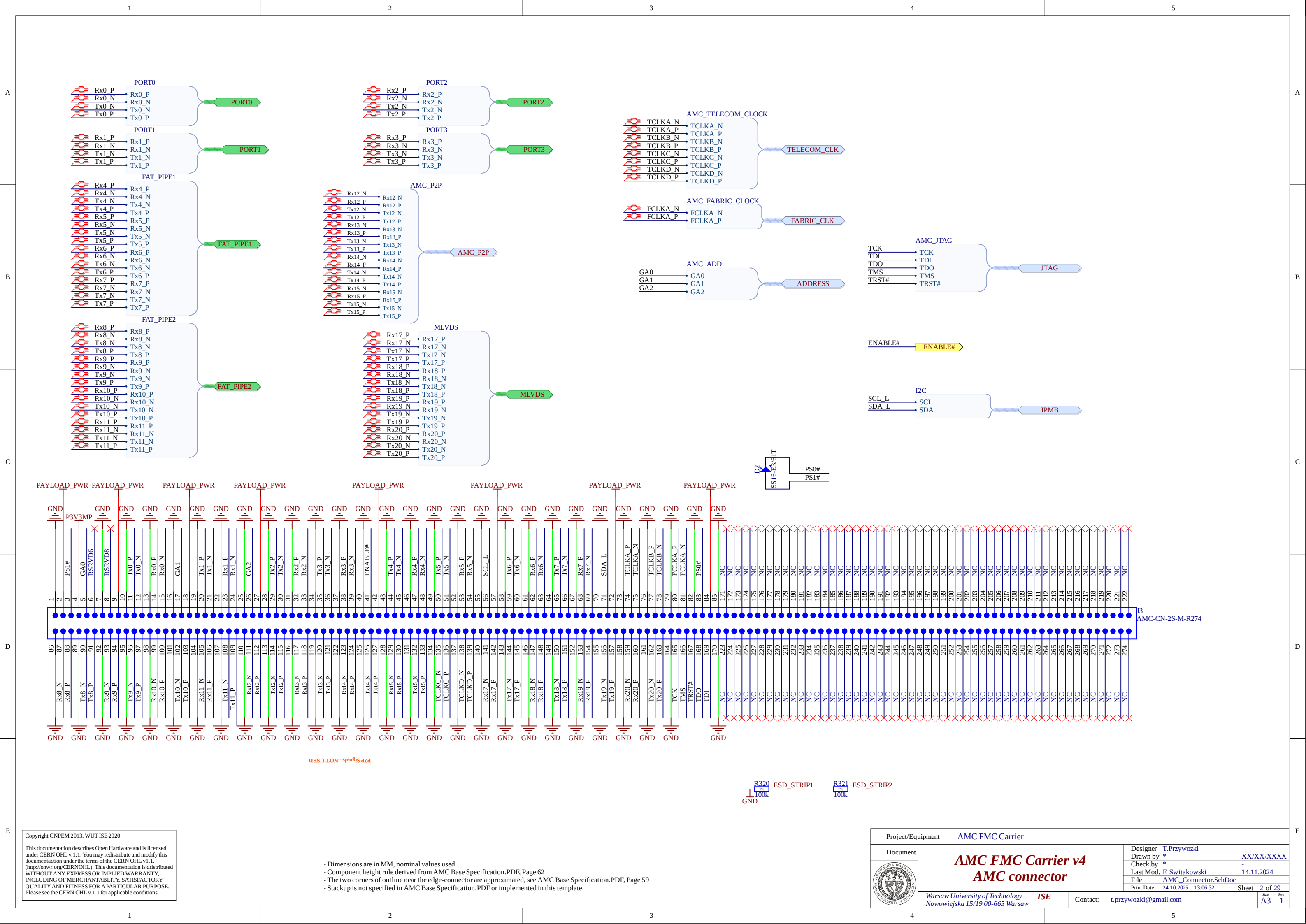
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Project/Equipment		AMC FMC Carrier	
Document		Designer T.Przywozki	
		Drawn by *	XX/XX/XXXX
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		Last Mod. M. Biernat	27.11.2024
		File	AMC_FMC_Carrier.SchDoc
		Print Date	24.10.2025 13:06:32
		Sheet	1 of 29
		Size	A3
		Rev	4.0.3
		Contact:	t.przywozki@gmail.com

AMC FMC Carrier v4
Top

Warsaw University of Technology
Nowowiejska 15/19 00-665 Warsaw

ISE

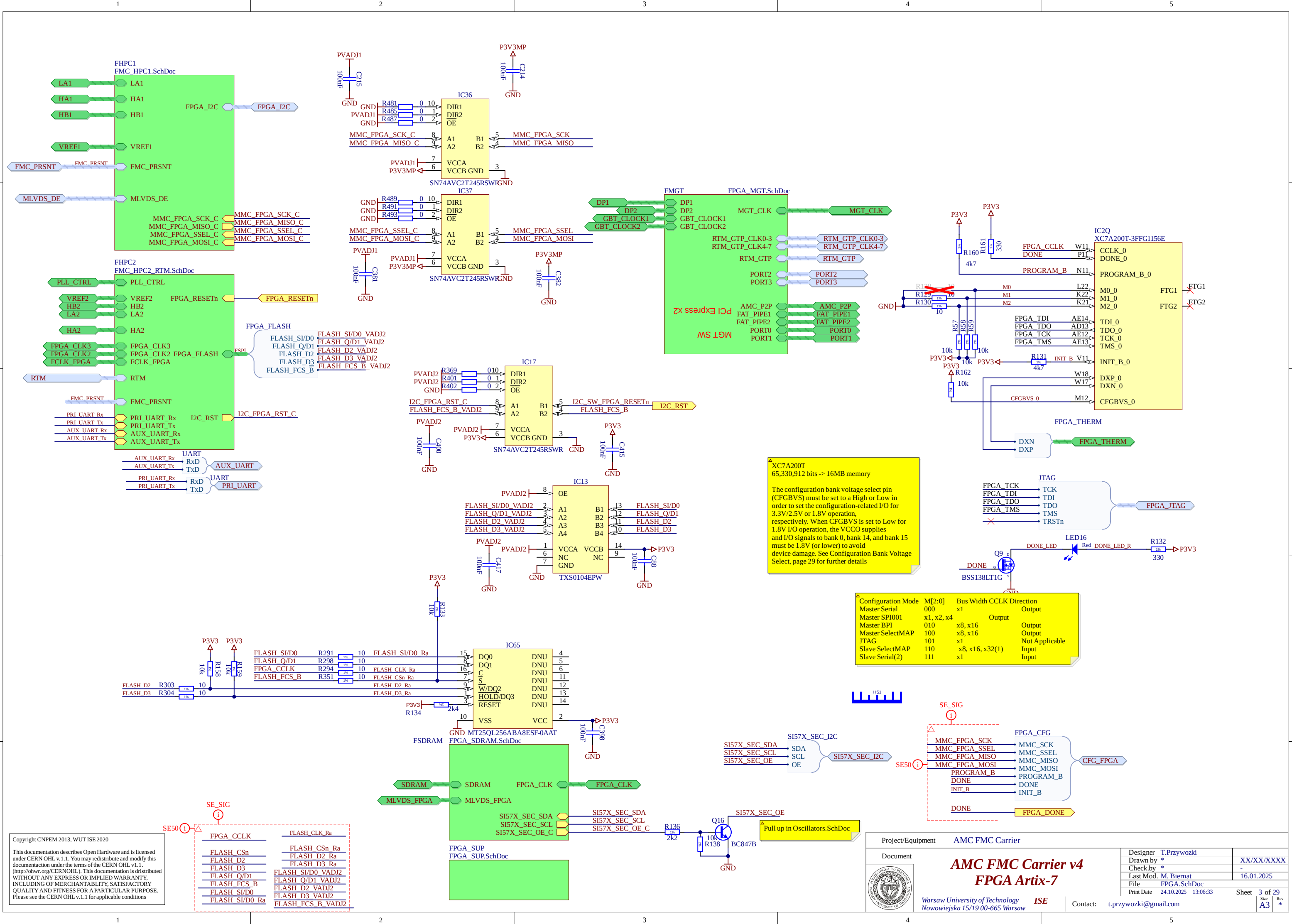


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- Dimensions are in MM, nominal values used
- Component height rule derived from AMC Base Specification.PDF, Page 62
- The two corners of outline near the edge-connector are approximated, see AMC Base Specification.PDF, Page 59
- Stackup is not specified in AMC Base Specification.PDF or implemented in this template.

Project/Equipment		AMC FMC Carrier	
Document		Designer T.Przywozki	
		Drawn by *	XX/XX/XXXX
		Check by *	-
		Last Mod. F. Switakowski	14.11.2024
		File AMC_Connector.SchDoc	
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Warsaw University of Technology Nowowiejska 15/19 00-665 Warsaw		Contact: t.przywozki@gmail.com	Size A3 Rev 1



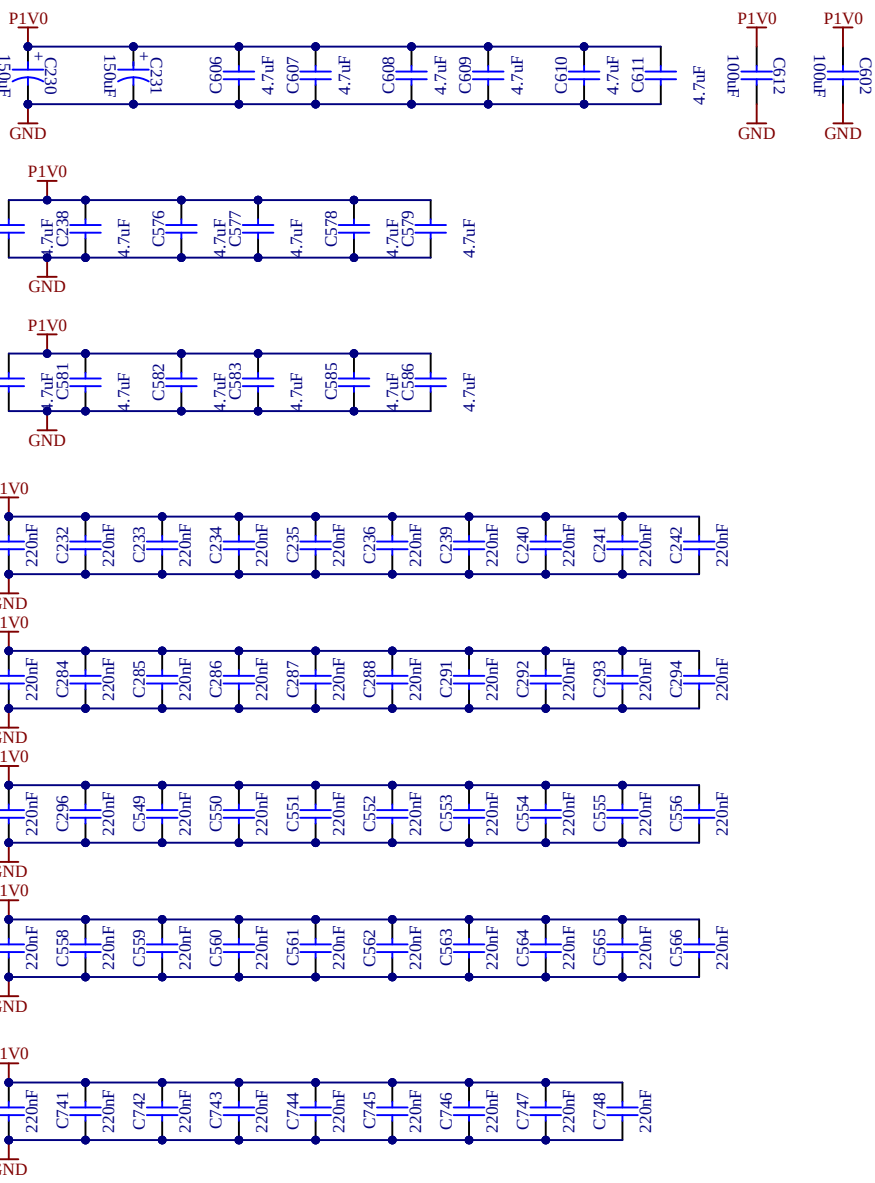
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VCCINT

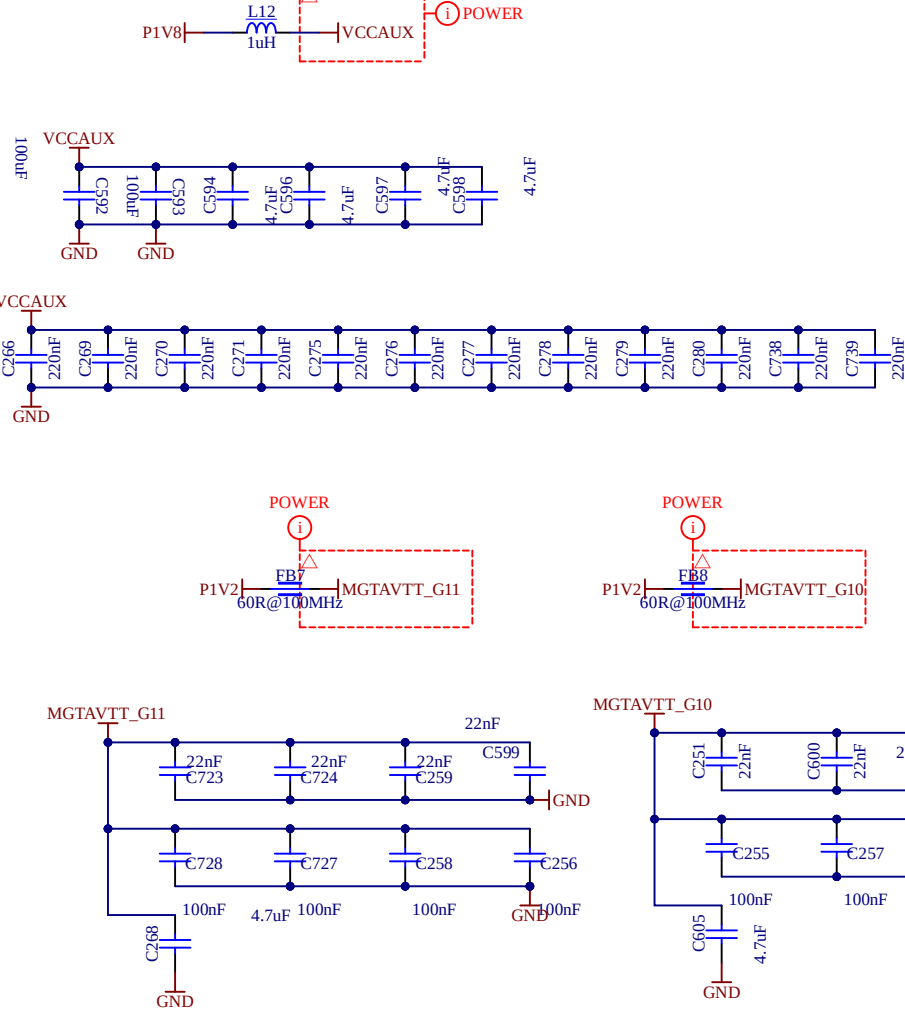
1x680uF, 2x100uF, 28x4.7uF

42x0.47uF on PG.34



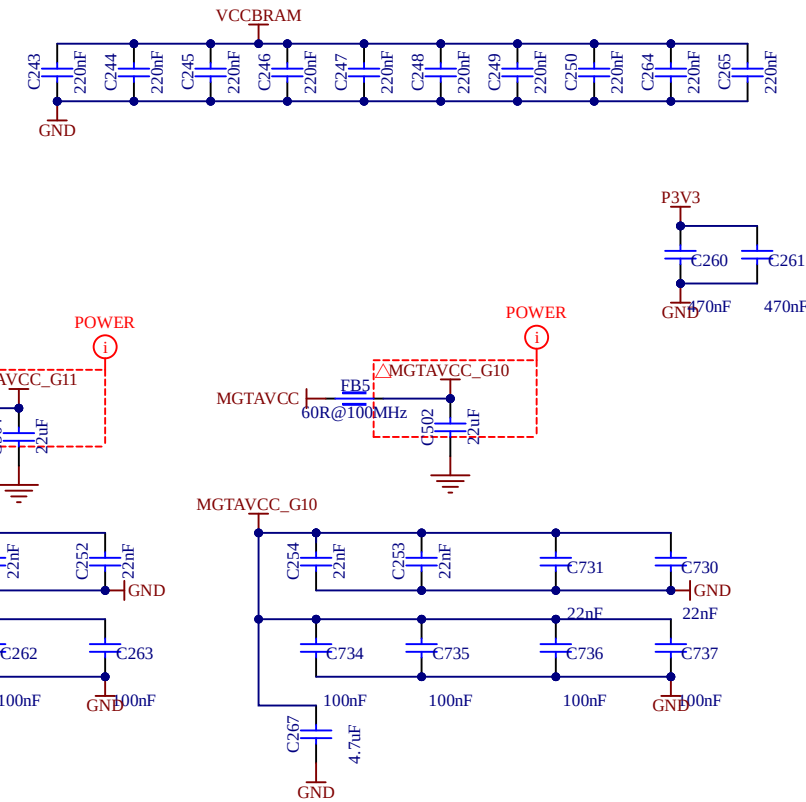
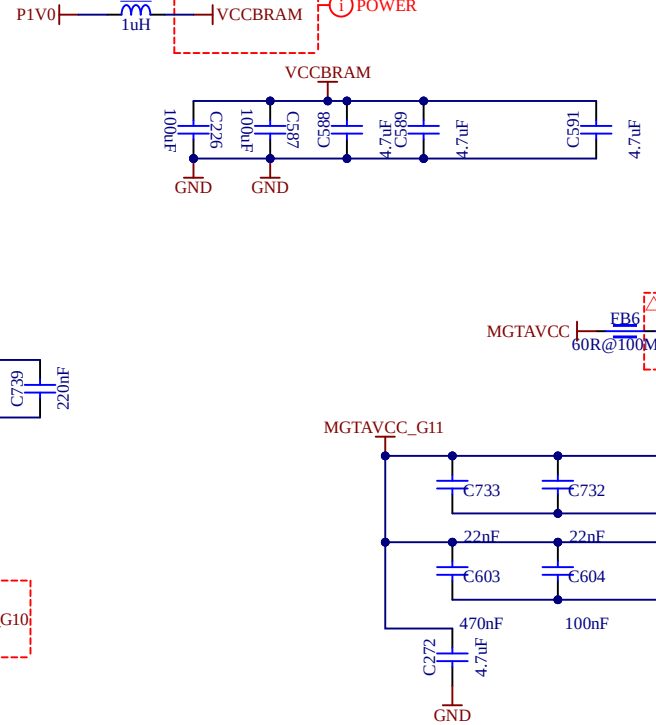
VCCAUX

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VCCBRAM

2x100uF, 4x4.7uF, 8x0.47uF



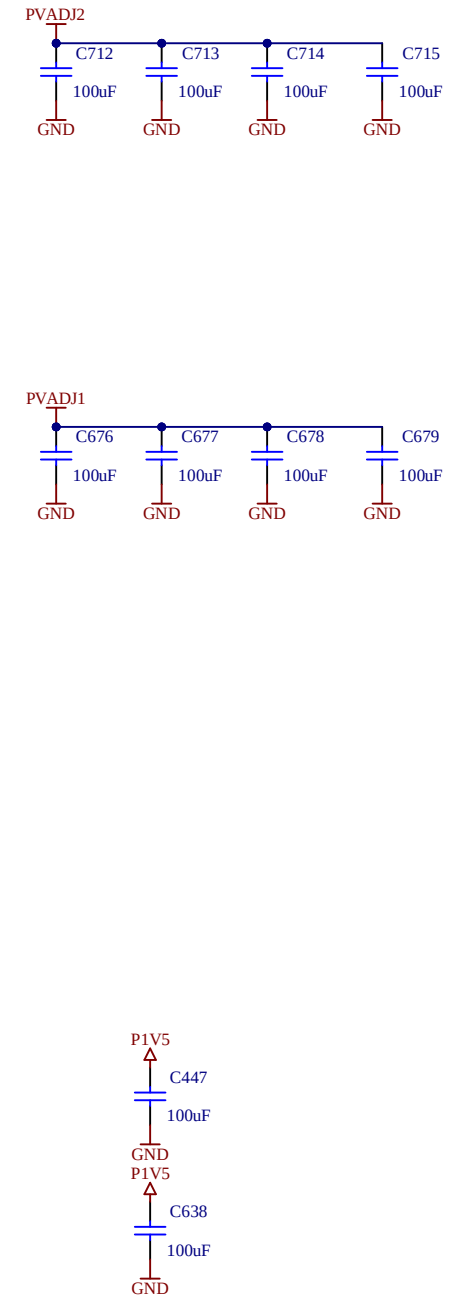
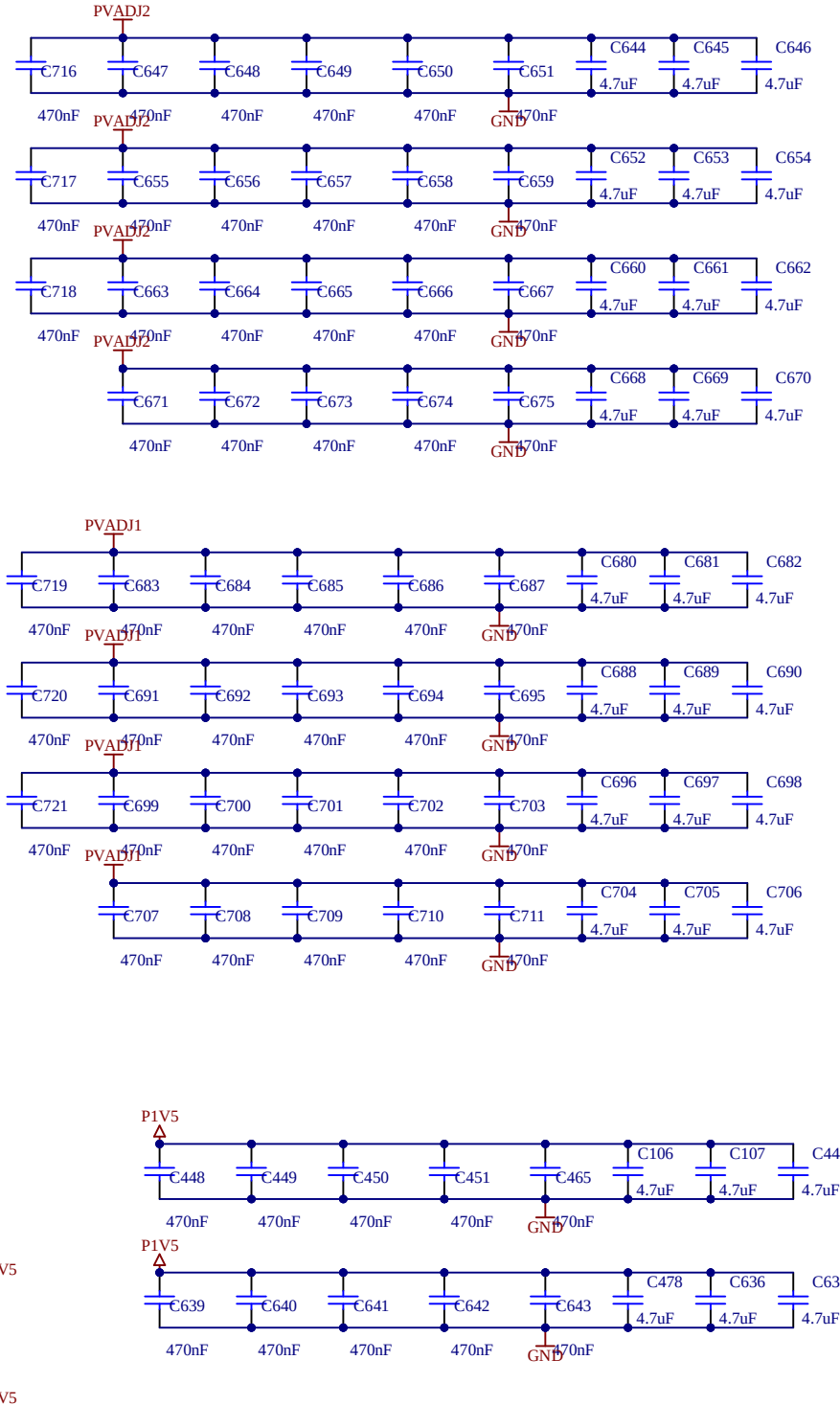
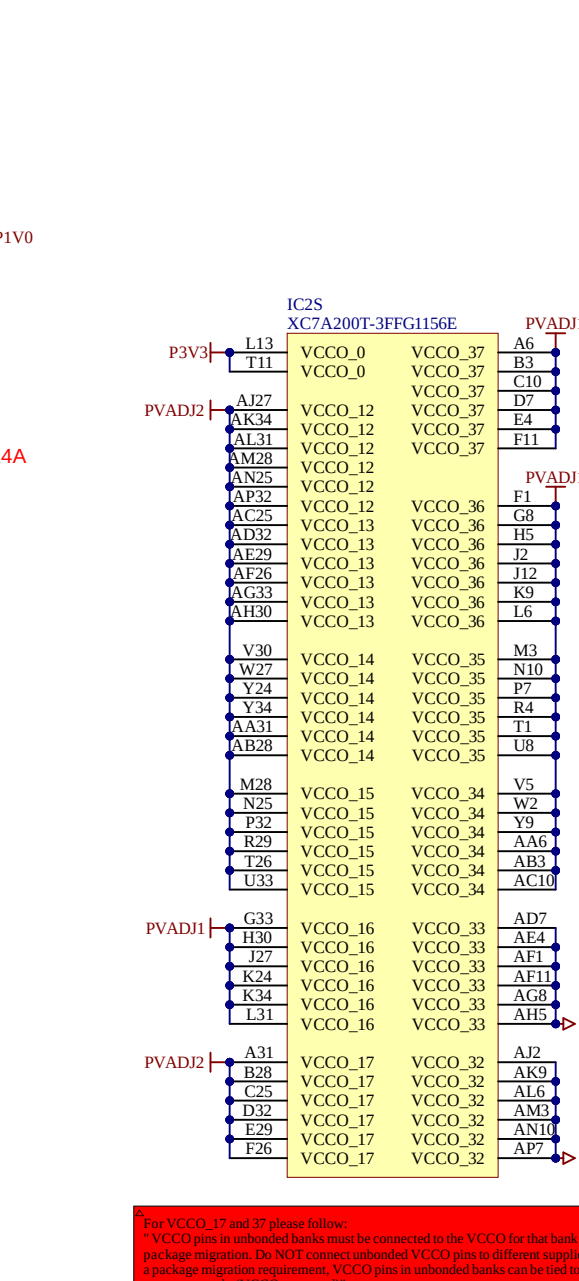
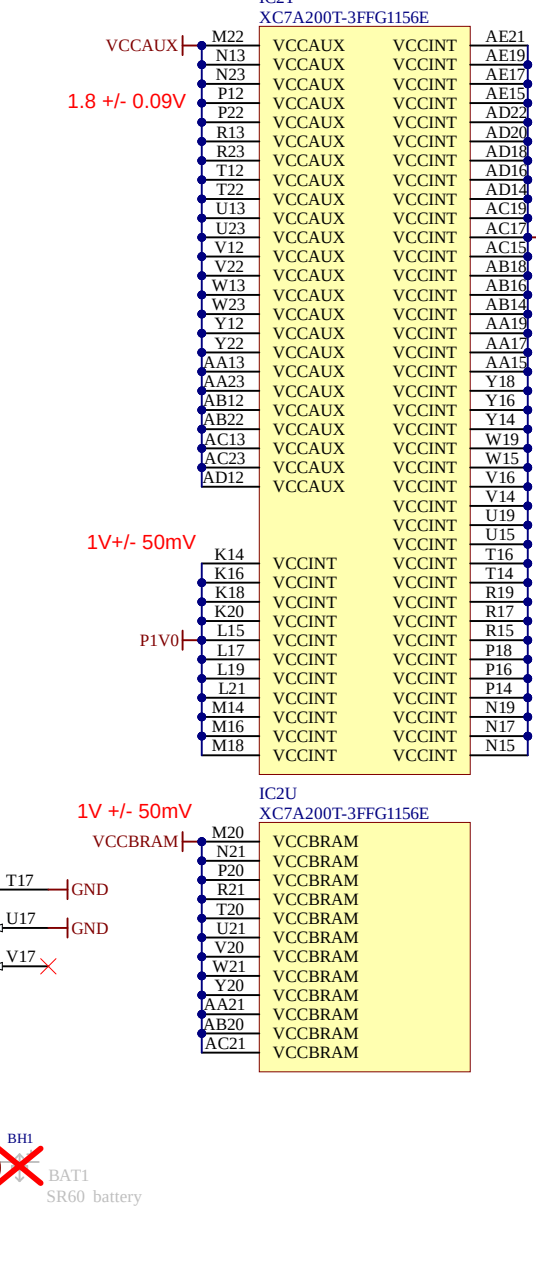
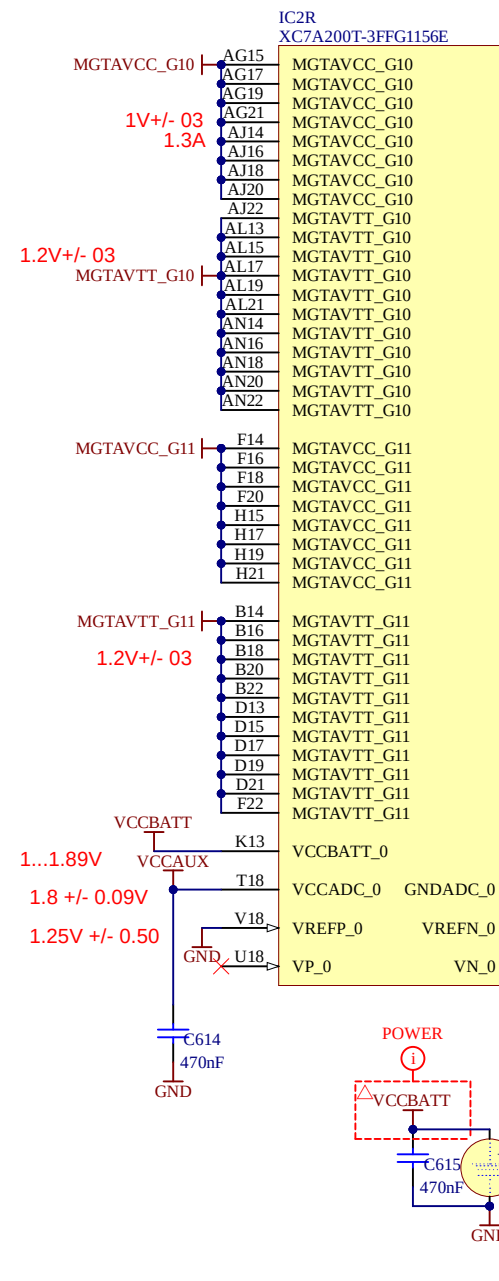
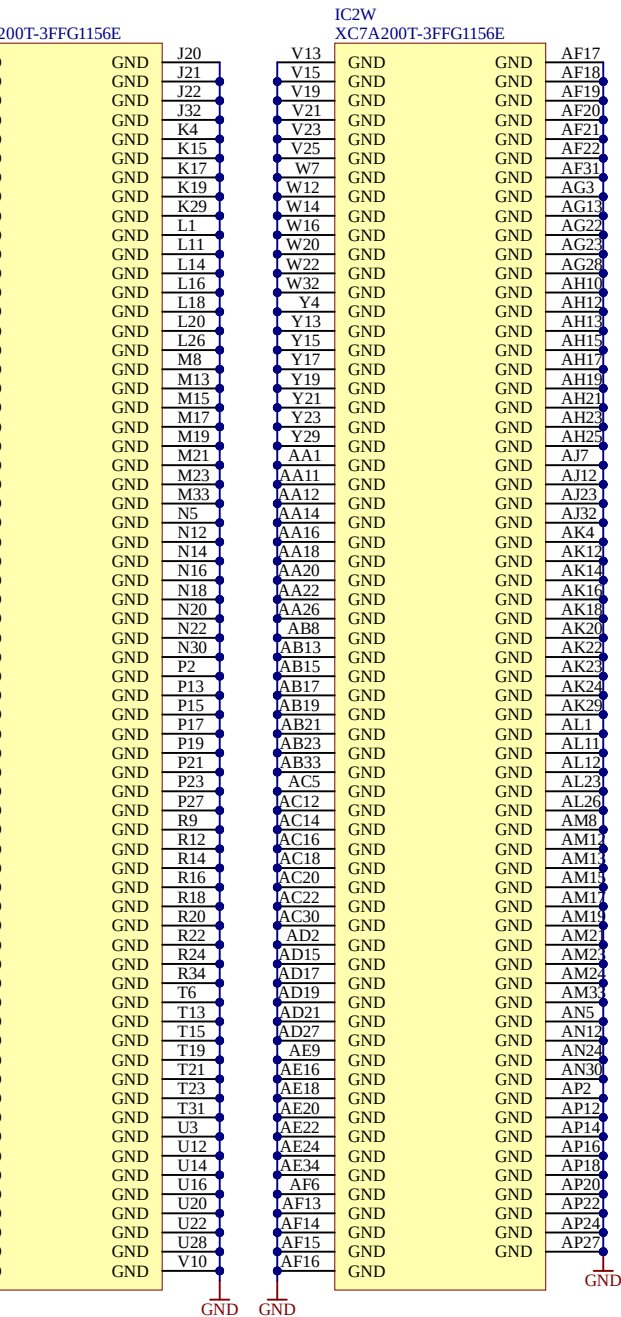
MGTAVCC	1V	2.3A		
MGTAVTT	1.2V	1.5A		
MGTVCCAUX	1.8V	0.5A		
VCCAUX	1.8V	1A		seq1
VCCAUX_IO	1.8/2V	0.1A		seq1
VCCBRAM	1V	1.8A		seq0
VCCINT	1V	6A		seq0
VCCO_1_2	1.2V	3.2A		seq2
VCCO_1_35	1.35V	0.9A		seq2
VCCO_1_5	1.5V	0.9A		seq2
VCCO_1_8	1.8V	0.9A		seq2
VCCO_2_5	2.5V	0.9A		seq2
VCCO_3_3	3.3V	0.9A		seq2

power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO voltage difference between VCCO and VCCAUX must not exceed 2.625V for longer than TVCCO2VCCAUX

power-on sequence to achieve minimum current draw for the GTP transceivers is VCCINT, VMGTAVCC, VCCINT, VMGTAVTT. There is no recommended sequencing for VMGTVCCAUX. Both VMGTAVCC and VCCINT can be ramped simultaneously. The recommended power-off sequence is the reverse of the power-on sequence to achieve minimum current draw.

When VMGTAVTT is powered before VMGTAVCC and VMGTAVTT - VMGTAVCC > 150 mV and VMGTAVCC < 0.7V, the VMGTAVTT current draw can increase by 460 mA per transceiver during VMGTAVCC ramp up. The duration of the current draw can be up to 0.3 x TMGTAVCC (ramp time from GND to 90% of VMGTAVCC). The reverse is true for power-down.

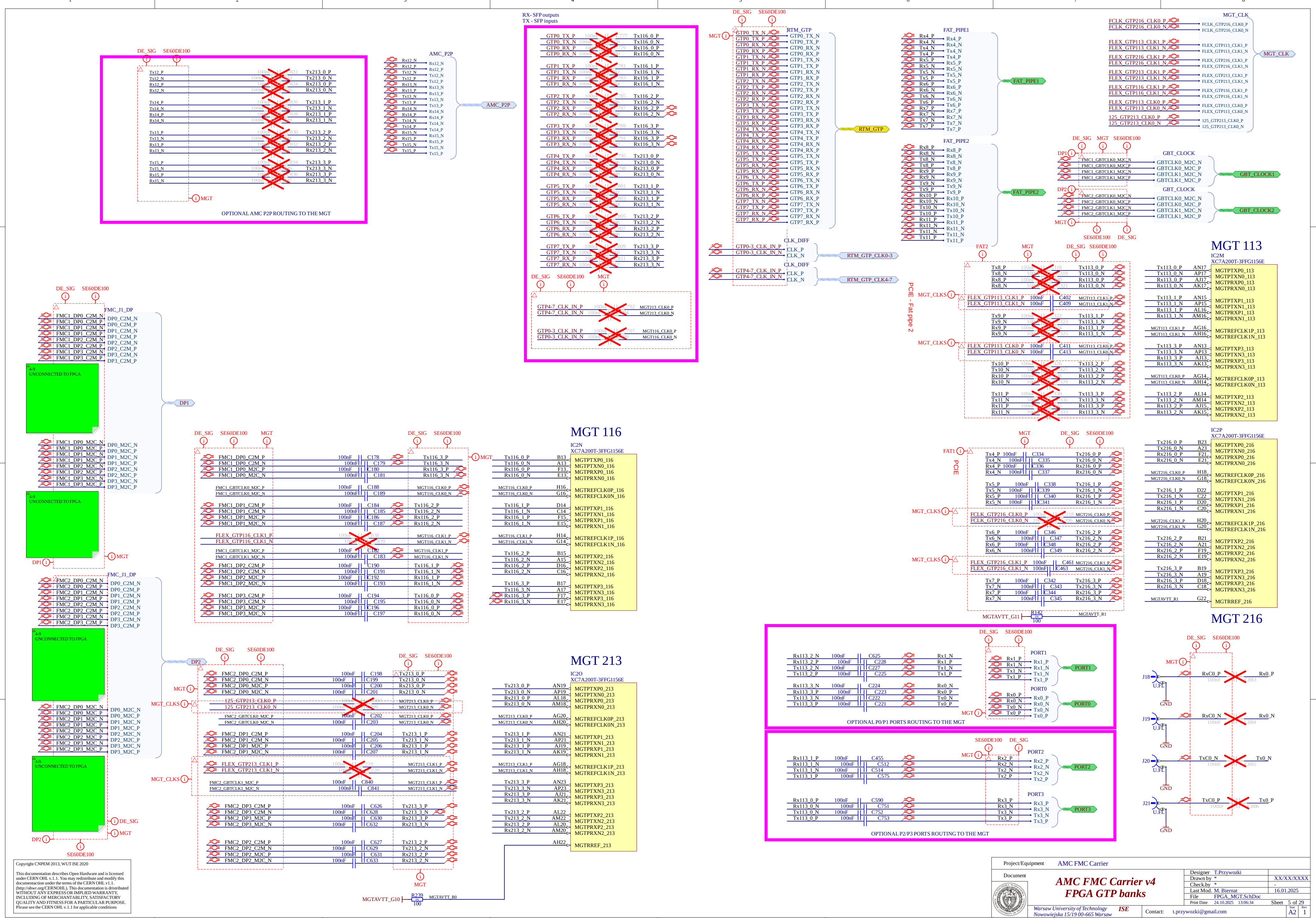
• When VMGTAVTT is powered before VCCINT and VMGTAVTT - VCCINT > 150 mV and VCCINT < 0.7V, the VMGTAVTT current draw can increase by 50 mA per transceiver during VCCINT ramp up. The duration of the current draw can be up to 0.3 x TVCCINT (ramp time from GND to 90% of VCCINT). The reverse is true for power-down.

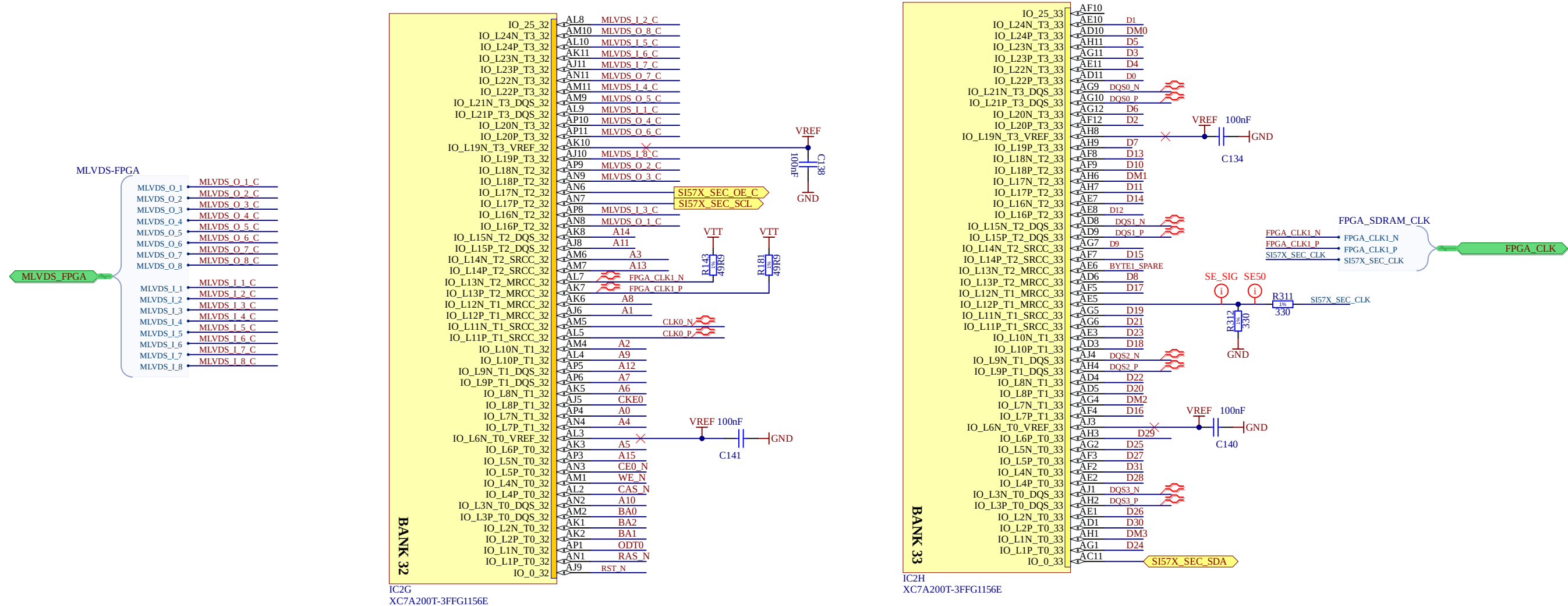


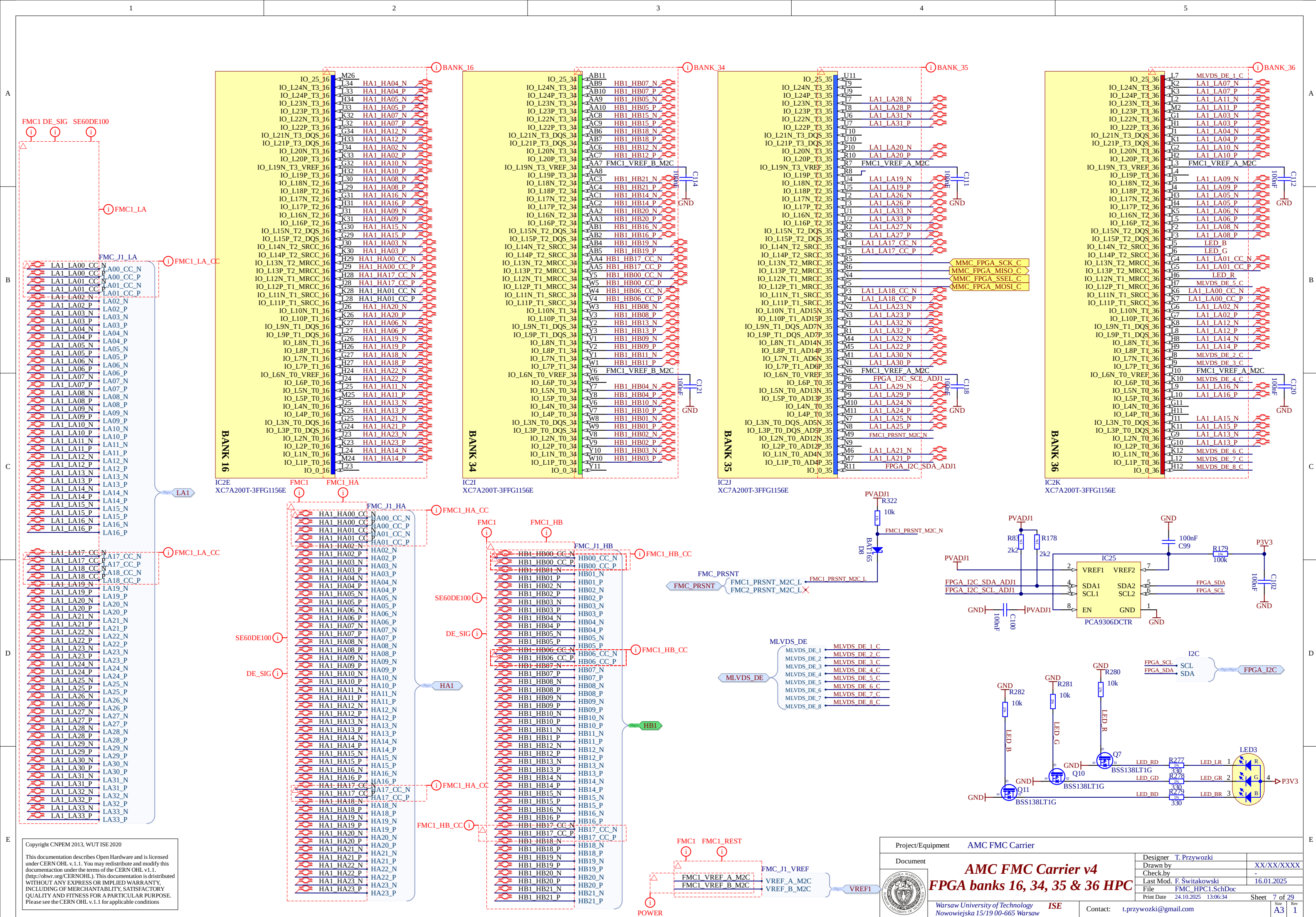
For VCCO_17 and T2 please follow:
- VCCO pins in unbonded banks must be connected to the VCCO for that bank for package migration. Do NOT connect unbonded VCCO pins to different supplies. Without package migration requirement, VCCO pins in unbonded banks can be tied to a common supply (VCCO or ground).
No migration available (since there is no 350T) - I recommend connecting to carry power rail and using as power decoupling placeholder for decaps. Remove bulk C222 and C219

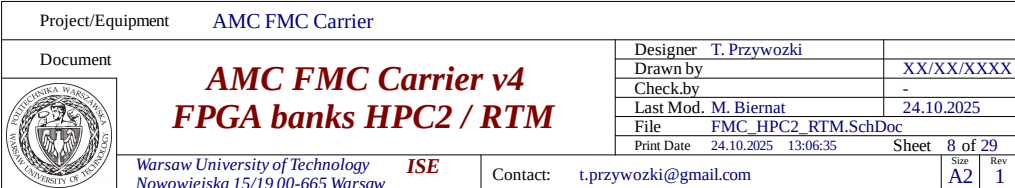
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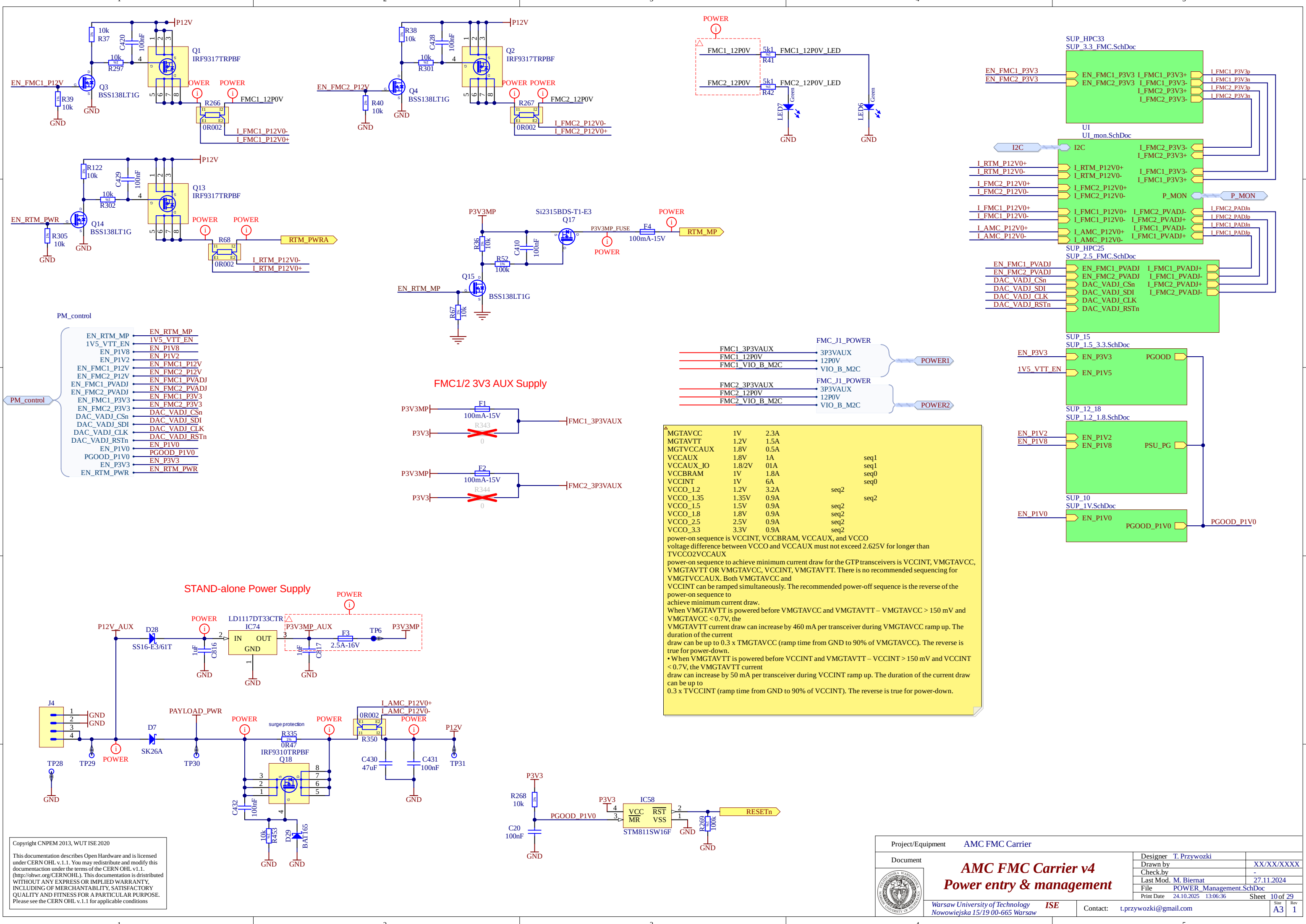
Project/Equipment	AMC FMC Carrier	Designer	T. Przywozki	
Document		Drawn by	*	XX/XX/XXXX
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		Last Mod.	F. Switkowski	27.11.2024
		File	FPGA_SUPSchDoc	
		Print Date	24.10.2025 13:06:33	Sheet 4 of 29
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Δ	MGTAVCC	1V	2.3A		
	MGTAVTT	1.2V	1.5A		
	MGTVCCAUX	1.8V	0.5A		
	VCCAUX	1.8V	1A	seq1	
	VCCAUX_IO	1.8/2V	01A	seq1	
	VCCBRAM	1V	1.8A	seq0	
	VCCINT	1V	6A	seq0	
	VCCO_1.2	1.2V	3.2A	seq2	
	VCCO_1.35	1.35V	0.9A	seq2	
	VCCO_1.5	1.5V	0.9A	seq2	
	VCCO_1.8	1.8V	0.9A	seq2	
	VCCO_2.5	2.5V	0.9A	seq2	
	VCCO_3.3	3.3V	0.9A	seq2	

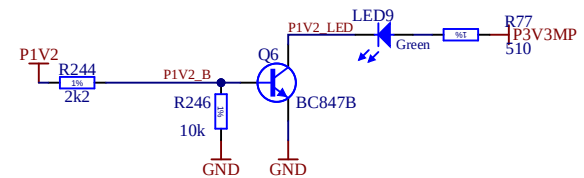
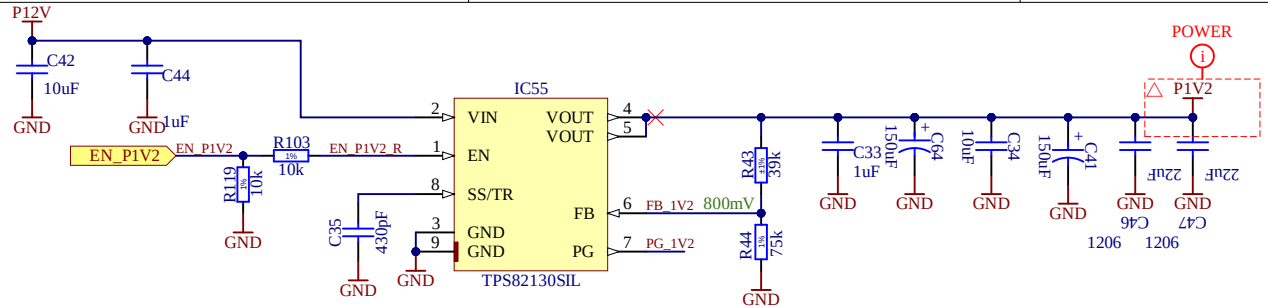
power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO voltage difference between VCCO and VCCAUX must not exceed 2.625V for longer than TVCCO2VCCAUX

power-on sequence to achieve minimum current draw for the GTP transceivers is VCCINT, VMGTAVCC, VMGTAVTT OR VMGTAVCC, VCCINT, VMGTAVTT. There is no recommended sequencing for VMGTVCCAUX. Both VMGTAVCC and VCCINT can be ramped simultaneously. The recommended power-off sequence is the reverse of the power-on sequence to achieve minimum current draw.

When VMGTAVTT is powered before VMGTAVCC and VMGTAVTT – VMGTAVCC > 150 mV and VMGTAVCC < 0.7V, the VMGTAVTT current draw can increase by 460 mA per transceiver during VMGTAVCC ramp up. The duration of the current draw can be up to 0.3 x TMGTAVCC (ramp time from GND to 90% of VMGTAVCC). The reverse is true for power-down.

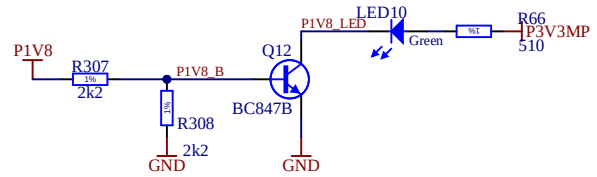
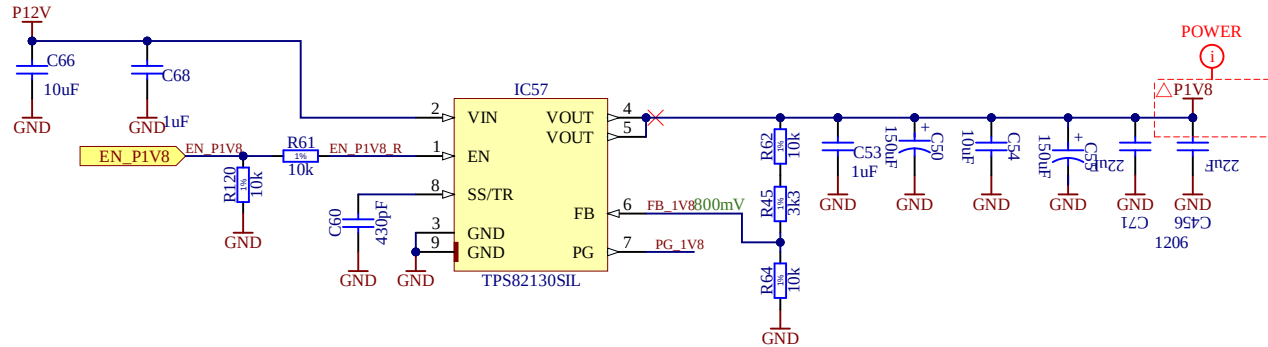
• When VMGTAVTT is powered before VCCINT and VMGTAVTT – VCCINT > 150 mV and VCCINT < 0.7V, the VMGTAVTT current draw can increase by 50 mA per transceiver during VCCINT ramp up. The duration of the current draw can be up to 0.3 x TVCCINT (ramp time from GND to 90% of VCCINT). The reverse is true for power-down.

A



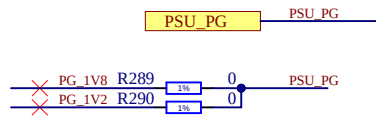
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B



B

C



C

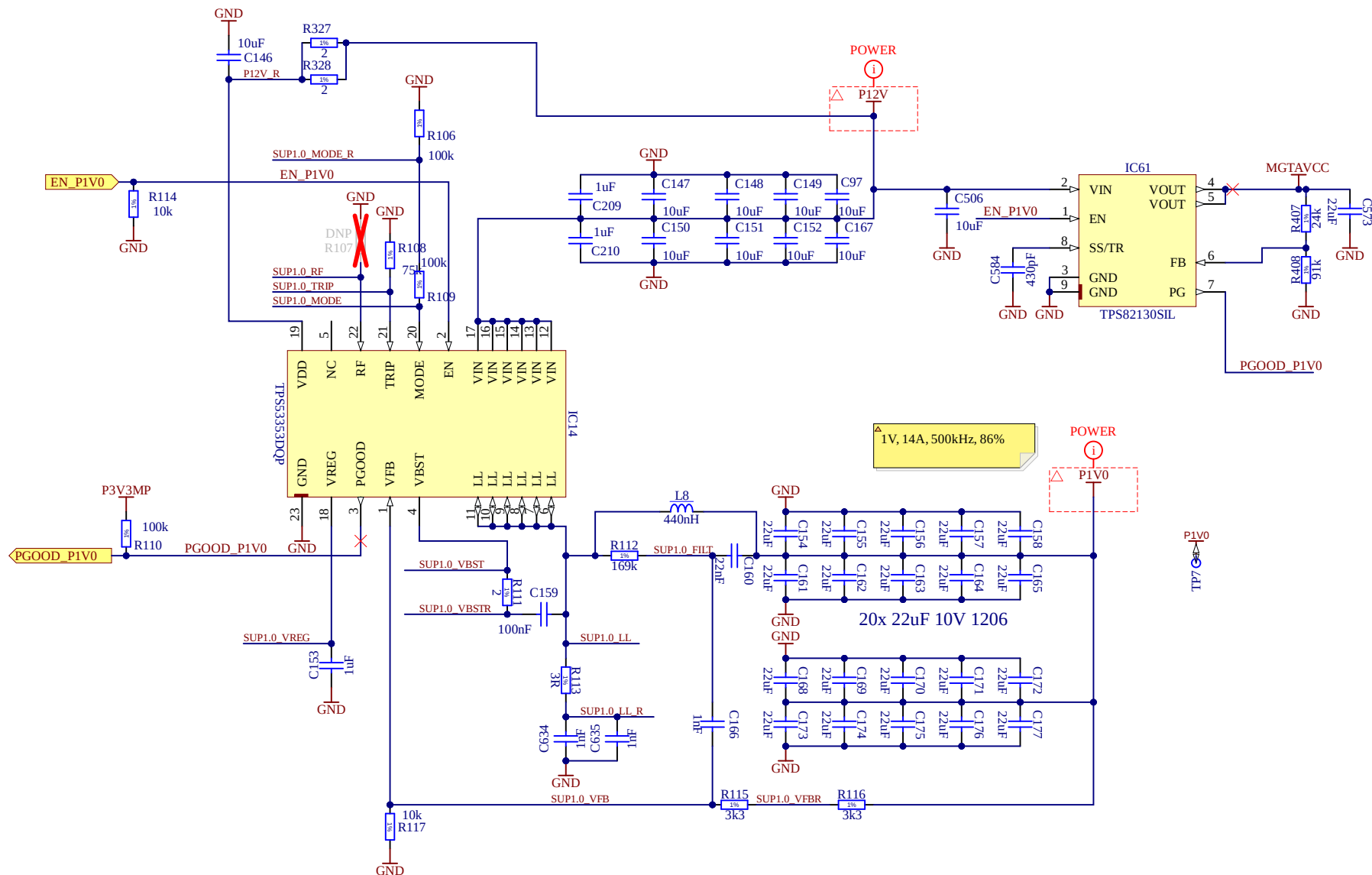
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	Document	AMC FMC Carrier v4 Power P1V2 & P1V8						
	Designer			T. Przywozki				
	Drawn by			XX/XX/XXXX				
	Check.by			-				
	Last Mod.			F. Świtakowski 27.11.2024				
	File			SUP_1.2_1.8.SchDoc				
	Print Date			24.10.2025 13:06:36				
Warsaw University of Technology Nowowiejska 15/19.00-665 Warsaw		ISE	Contact:	t.przywozki@gmail.com	Sheet	11 of 29	Size	Rev
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Project/Equipment **AMC FMC Carrier**

Document



AMC FMC Carrier v4 Power P1V0

Warsaw University of Technology
Nowowiejska 15/19 00-665 Warsaw

ISE

Contact: t.przywozki@gmail.com

Designer	T. Przywozki
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Drawn by

Check.by

Last Mod. F. Świtakows

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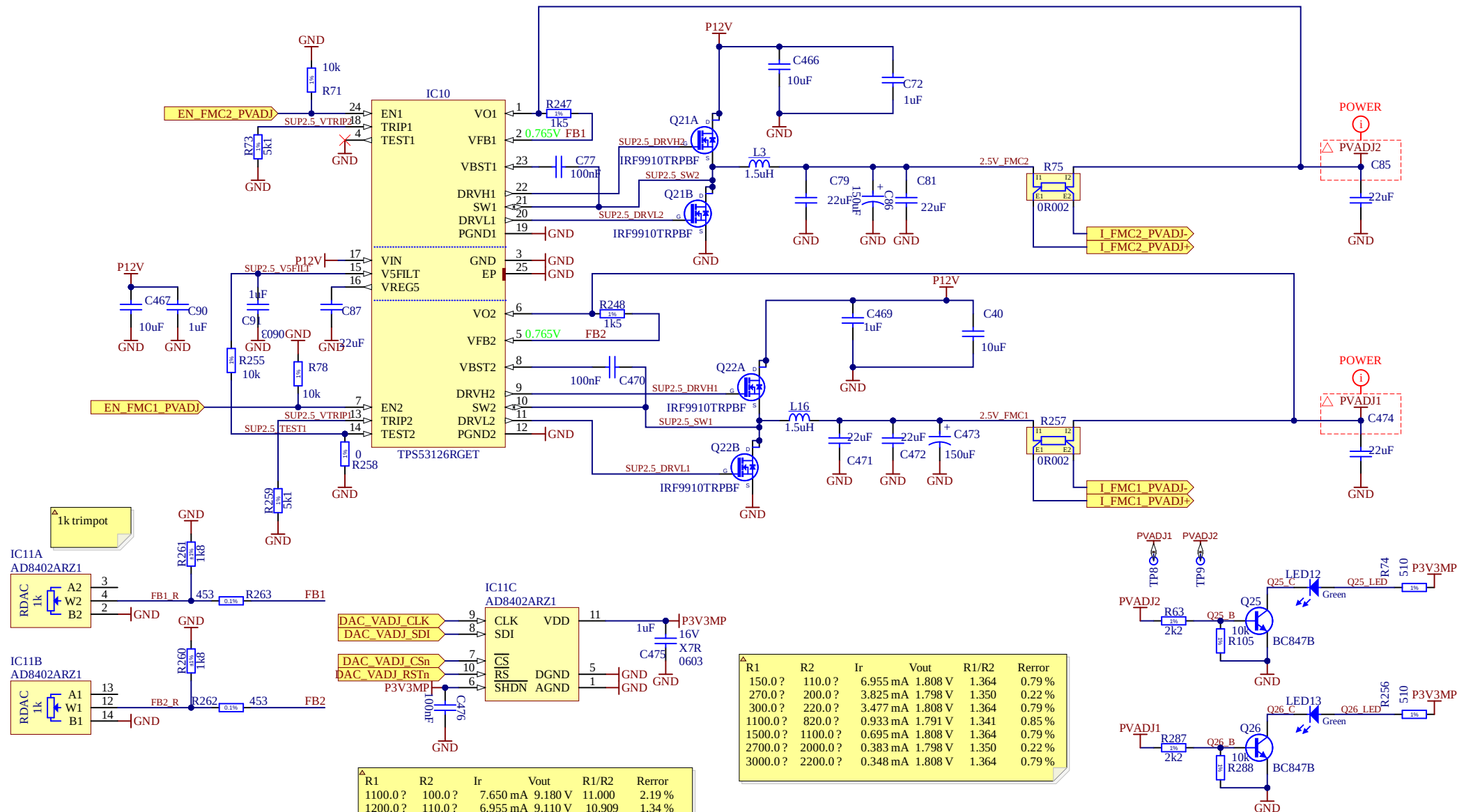
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Project/Equipment AMC FMC Carrier

Document

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Nowowiejska 15/19 00-665 Warsaw

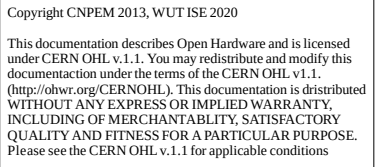
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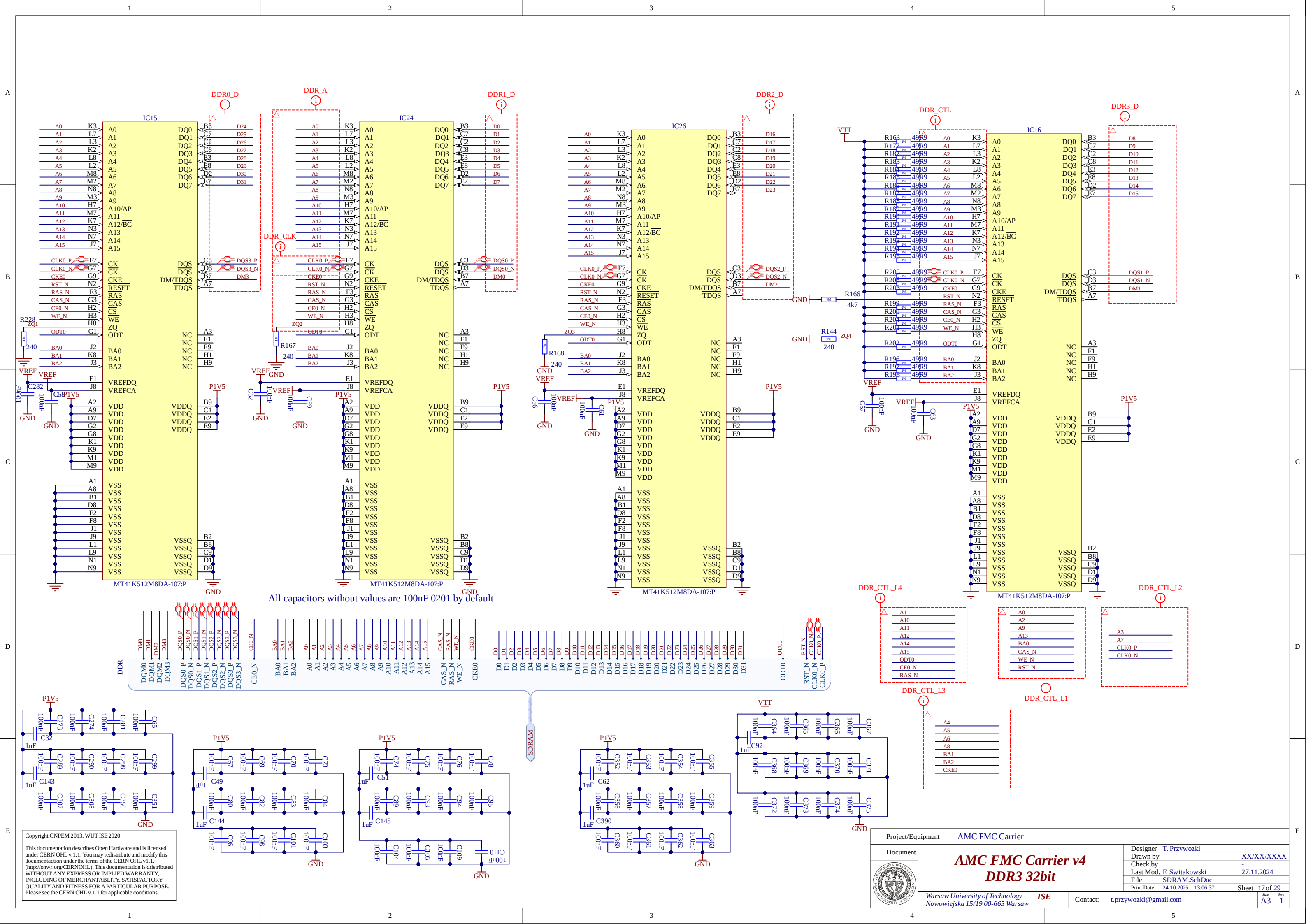
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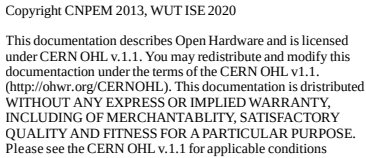
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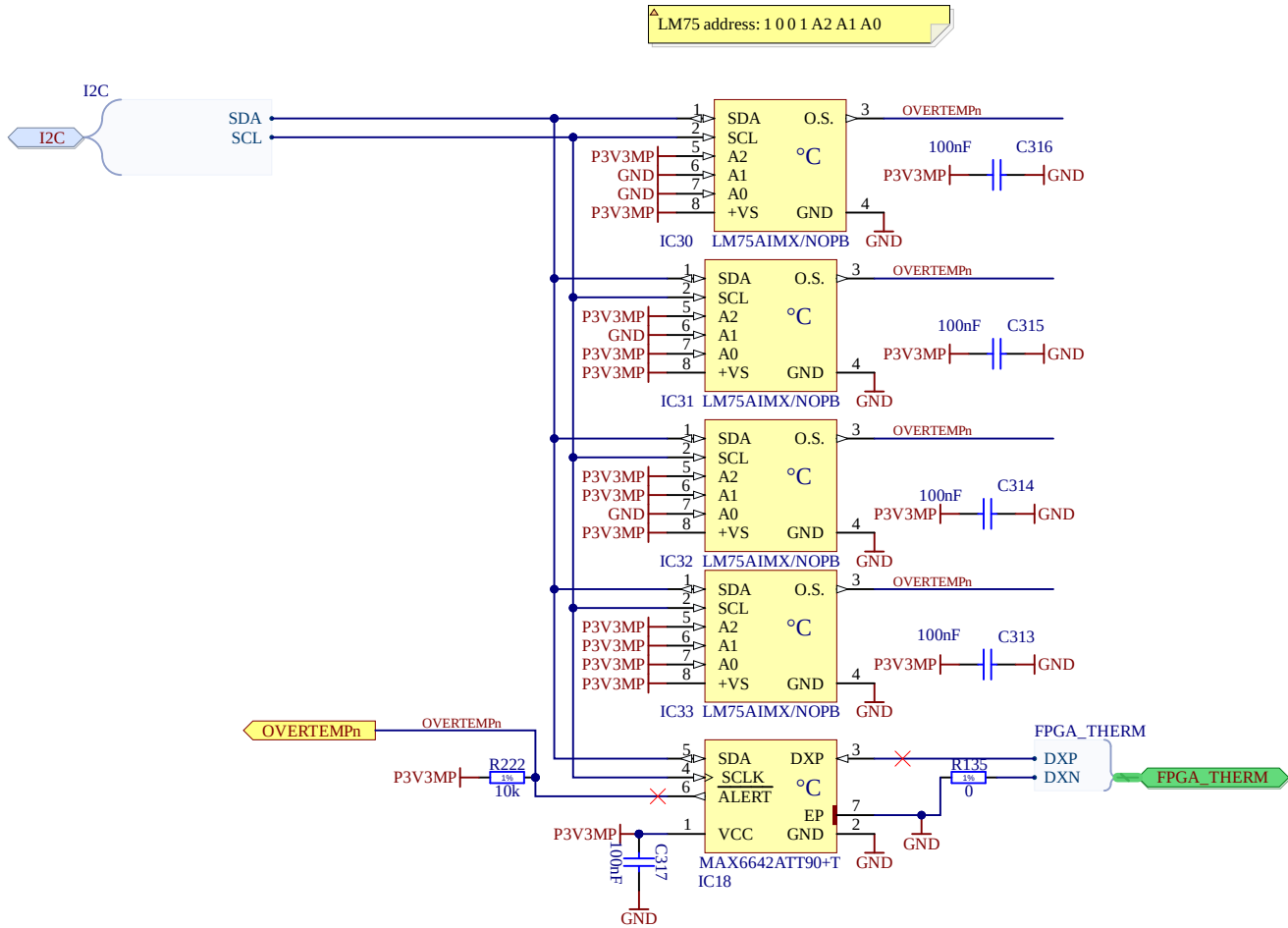
Contact: t.przywozki@gmail.com

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Project/Equipment AMC FMC Carrier

Document



AMC FMC Carrier v4 Thermometers

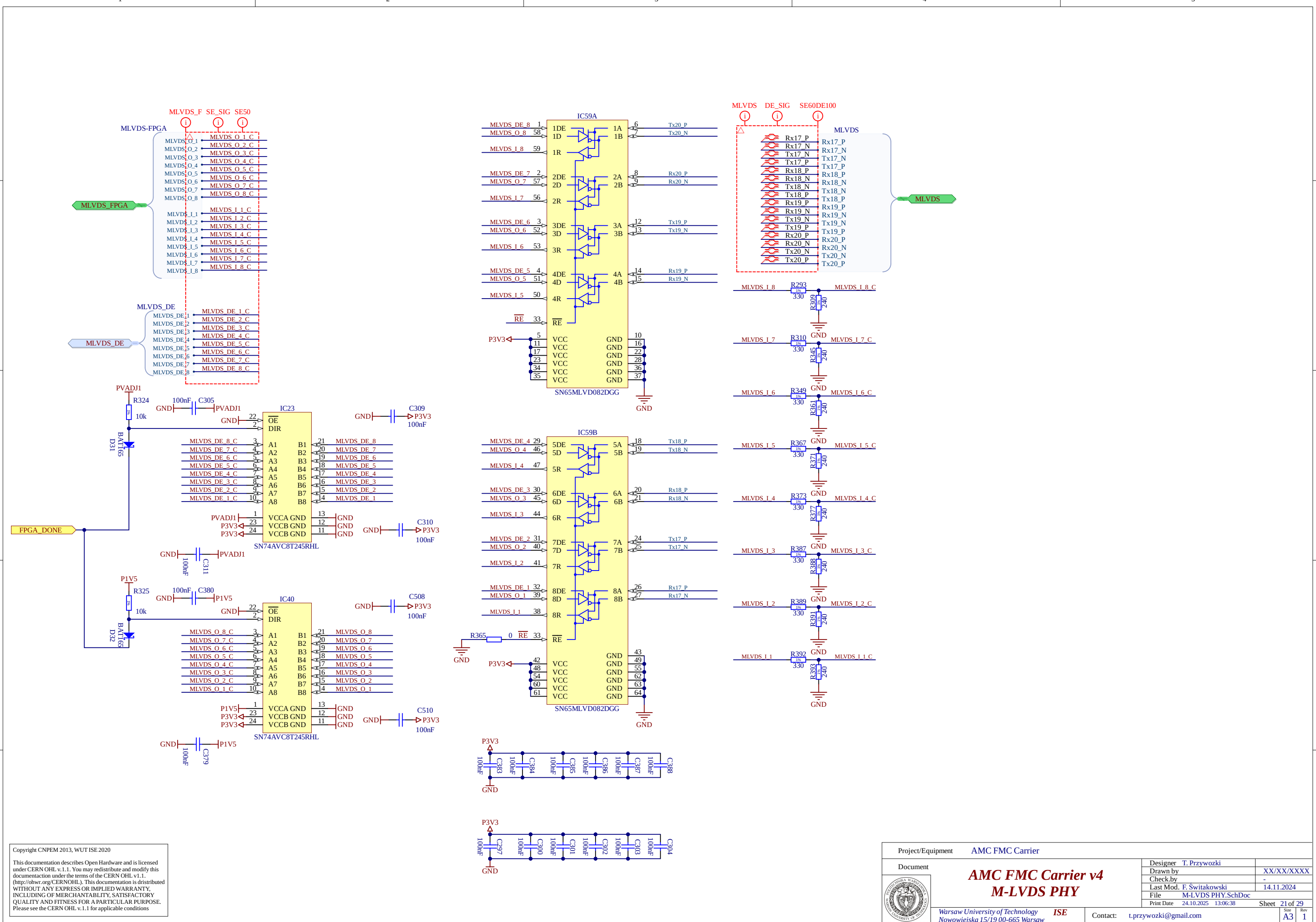
Warsaw University of Technology
Nowowiejska 15/19 00-665 Warsaw

ISE

Contact: t.przywozki@gmail.com

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Drawn by		XX/XX/XXXX
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Last Mod.	F. Świtkowski	14.11.2024
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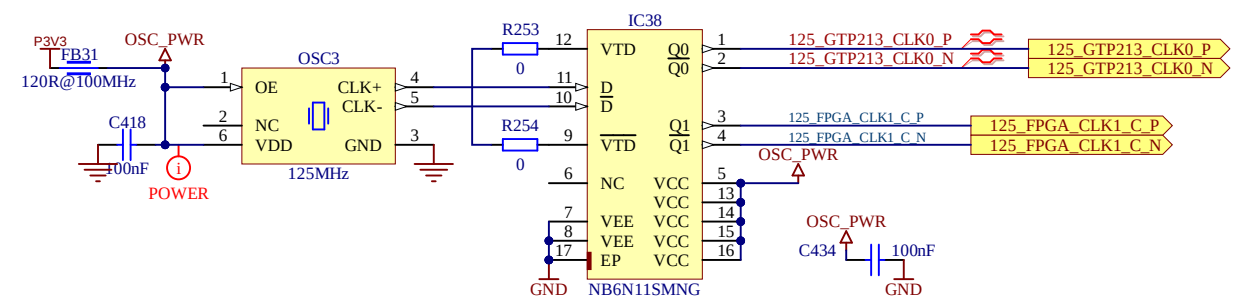
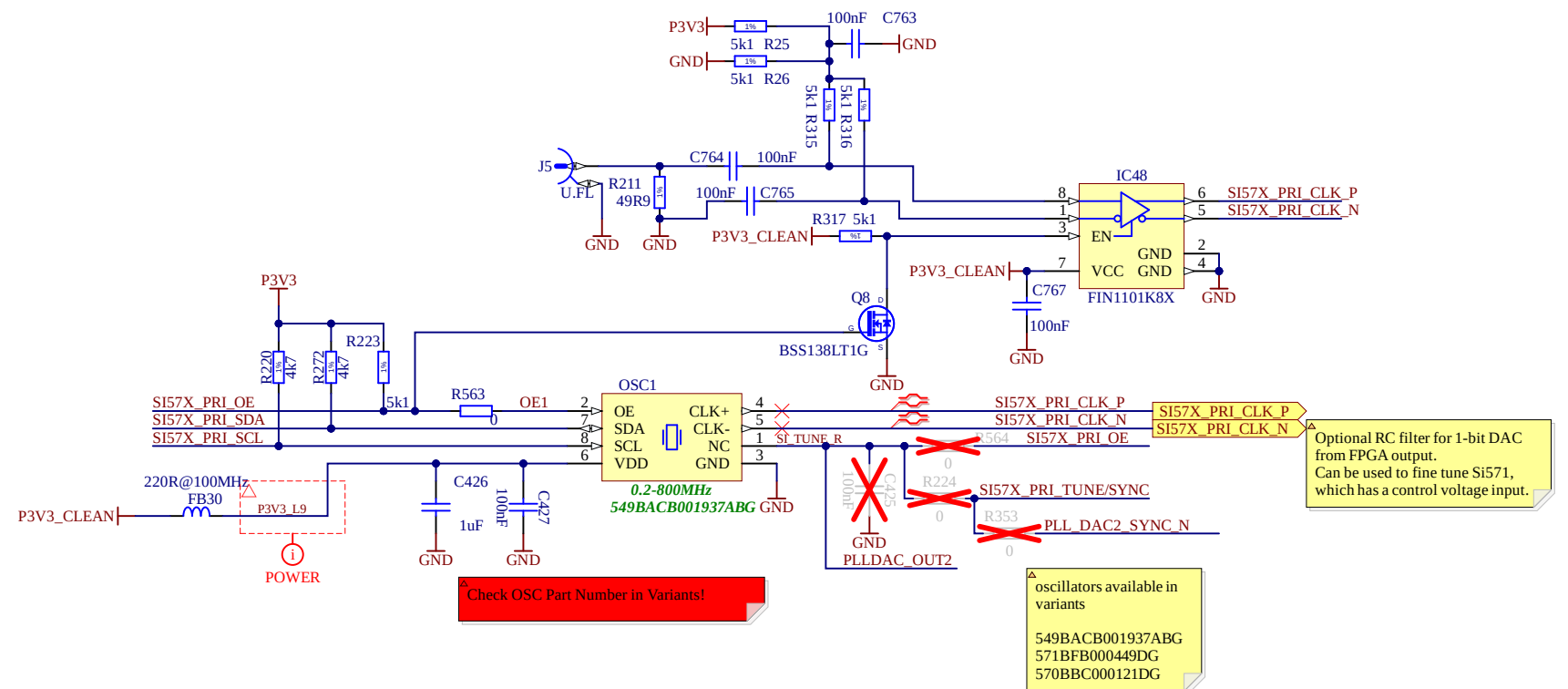
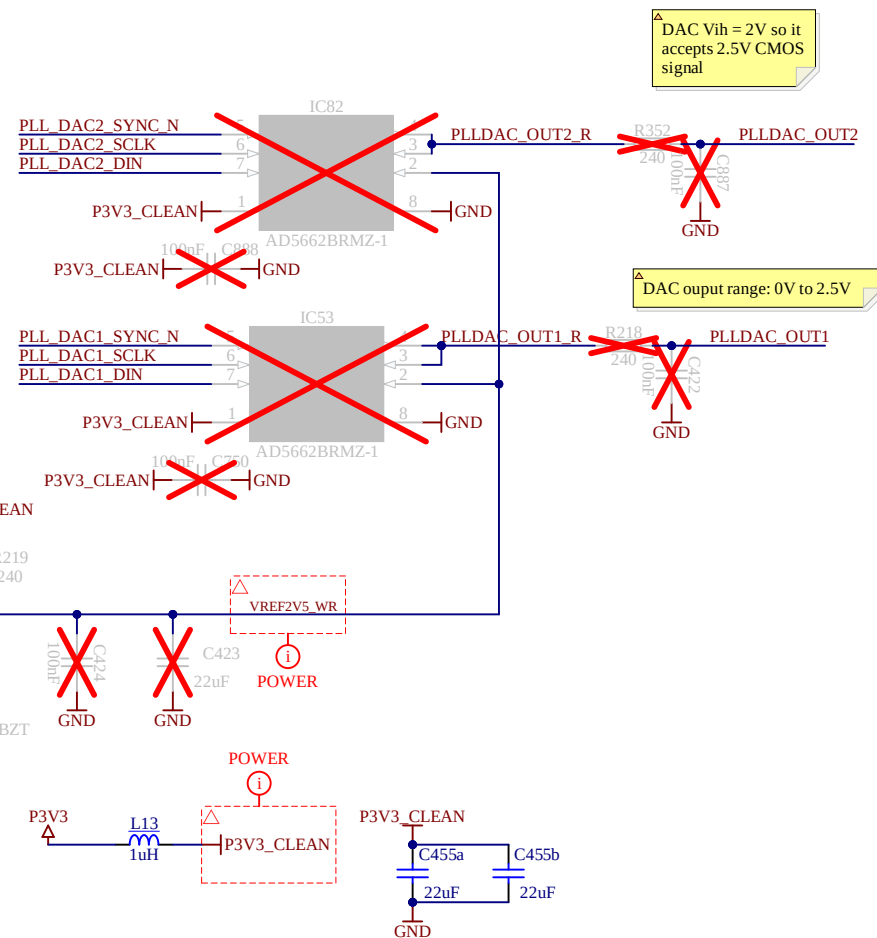
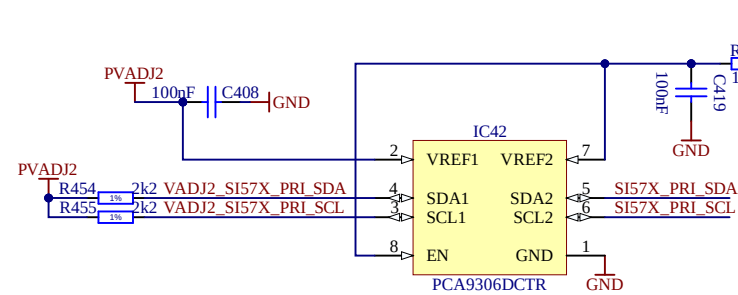
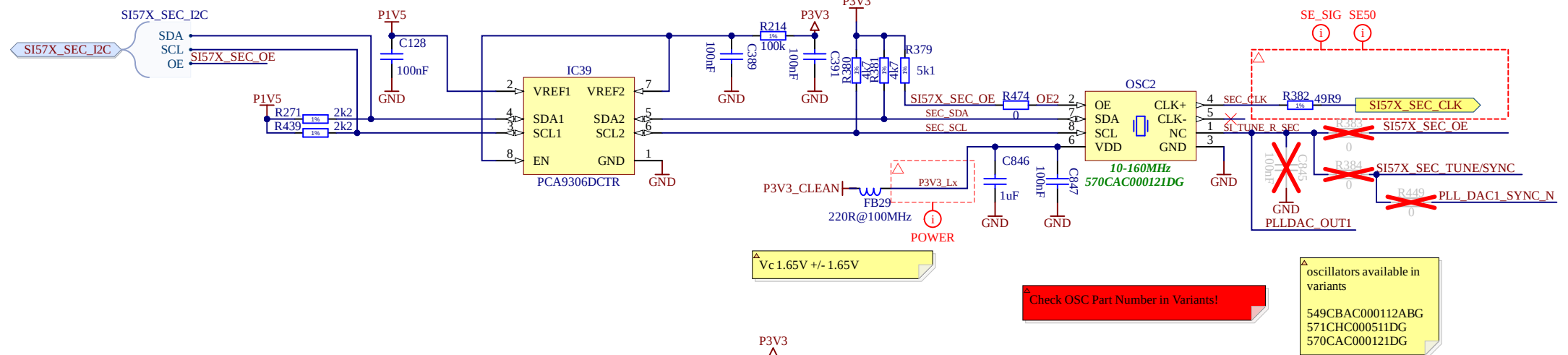
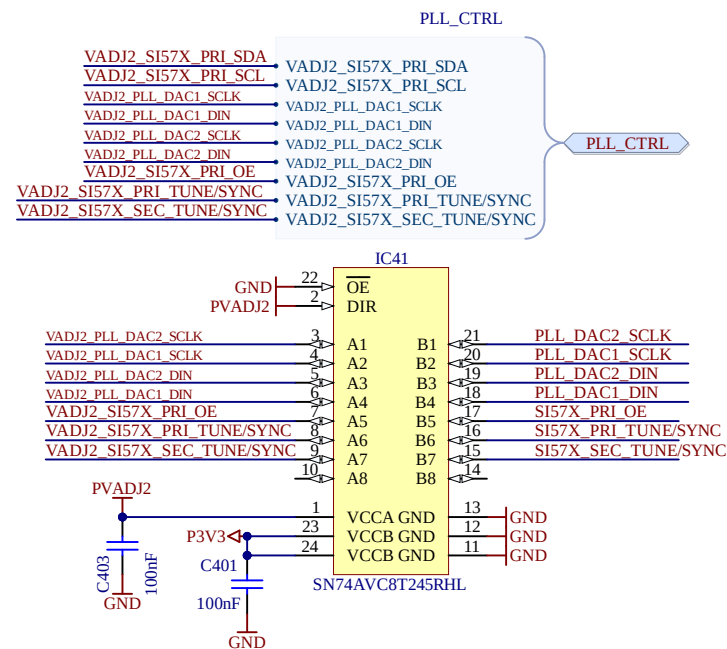
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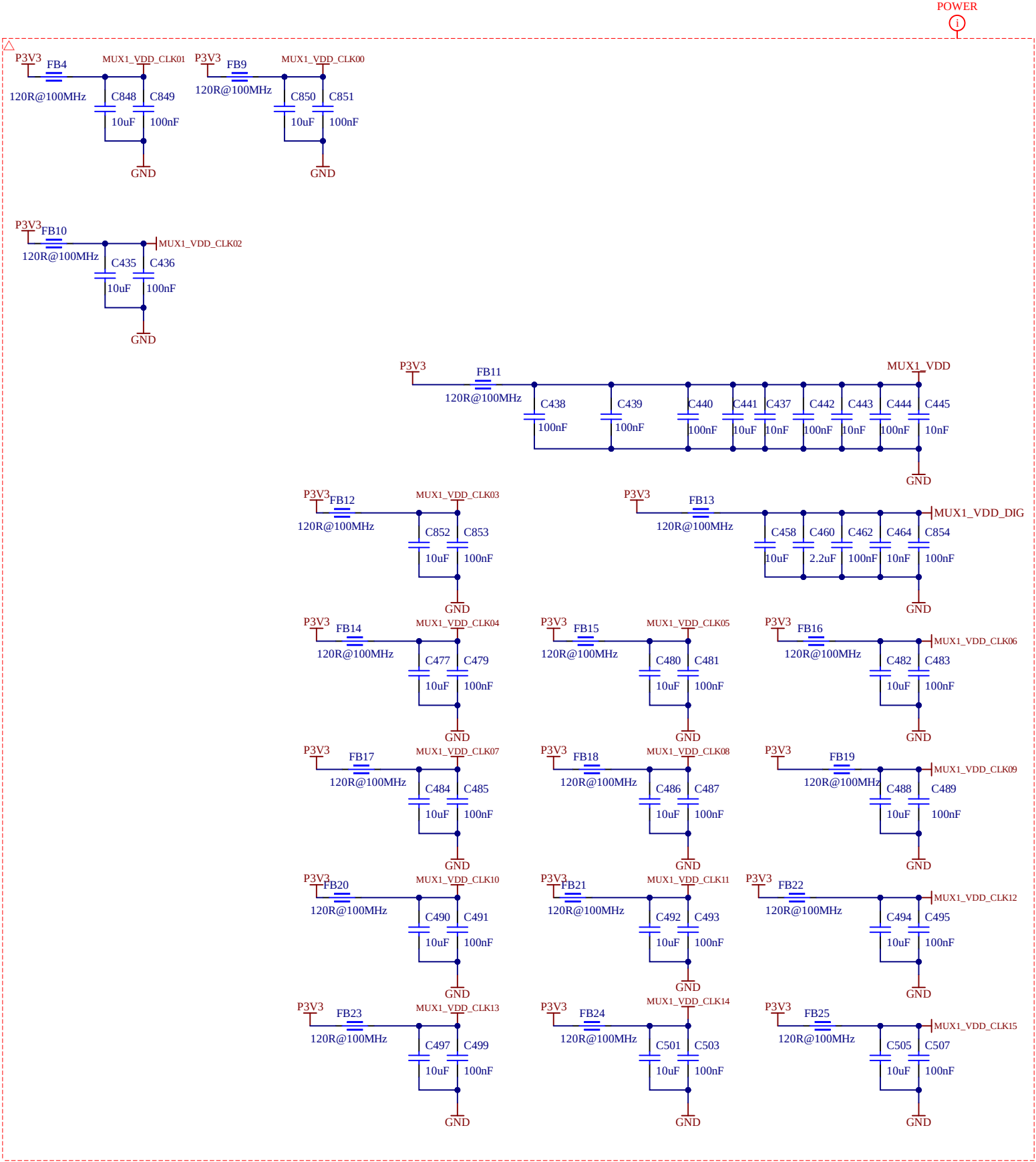


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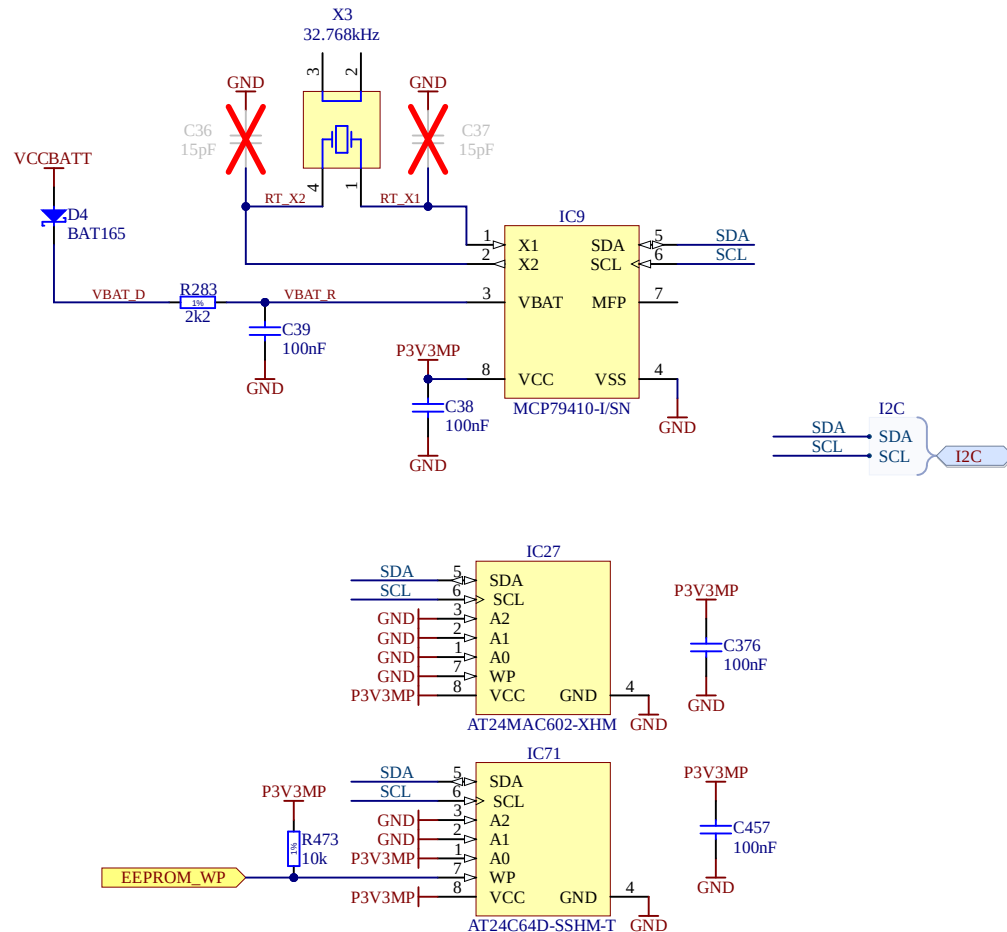




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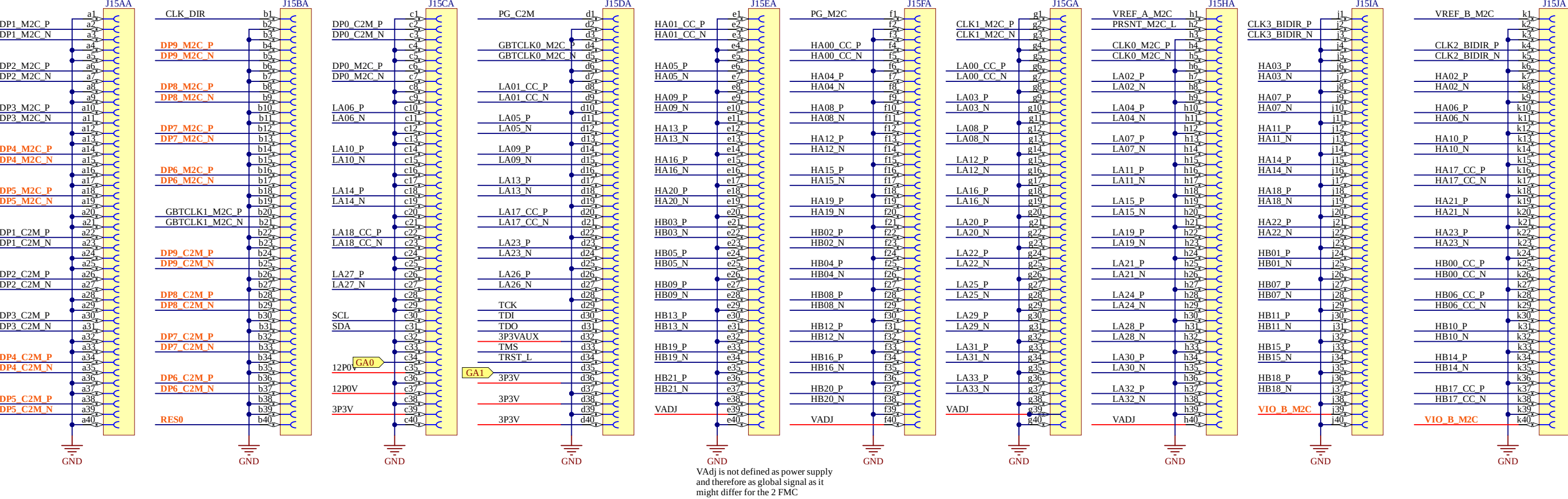
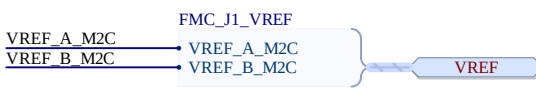
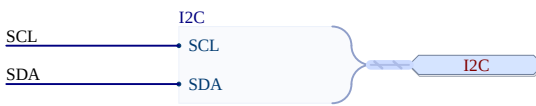
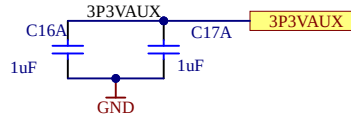
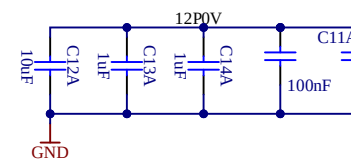
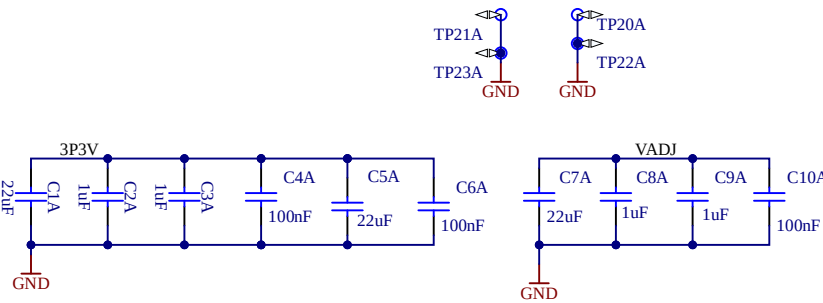
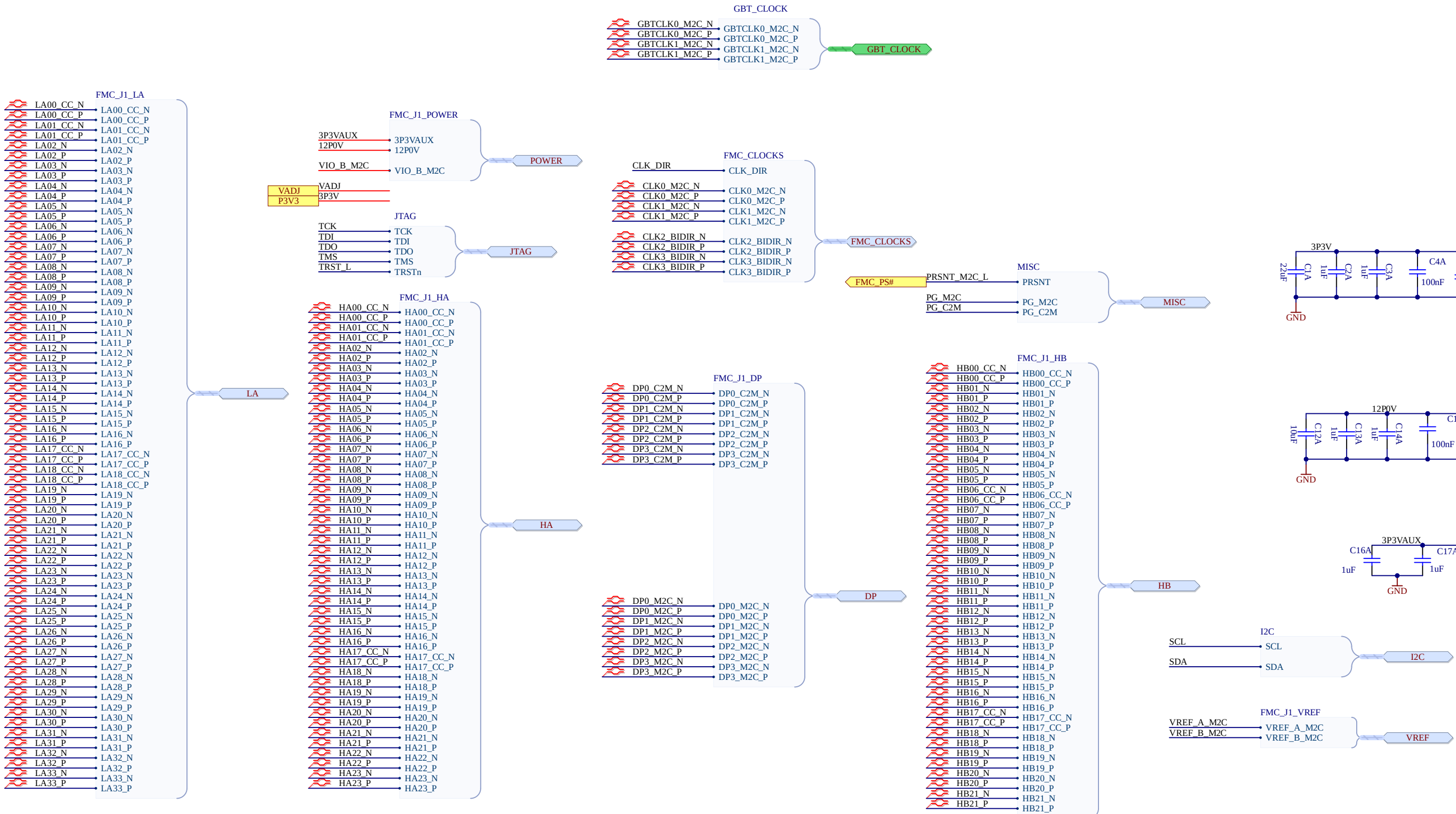
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Contact:		t.przywozki@gmail.com			
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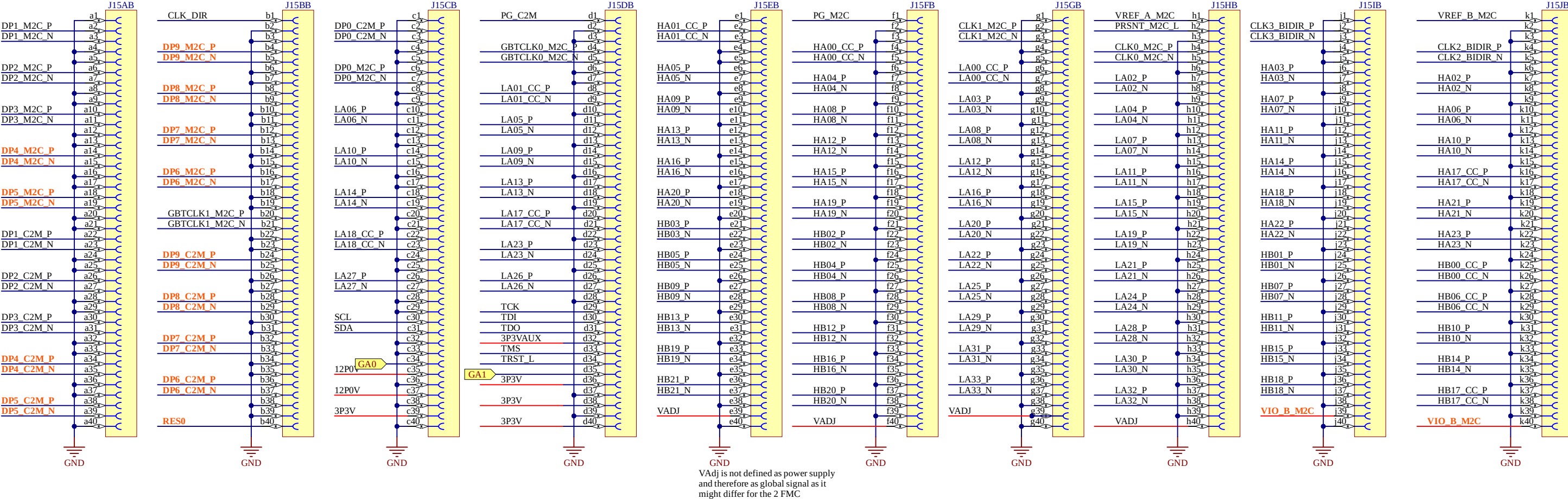
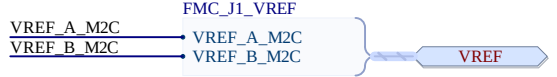
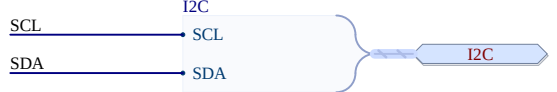
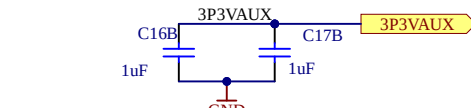
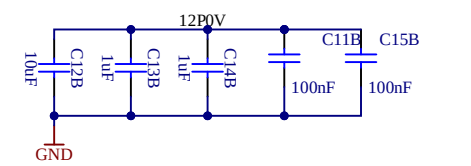
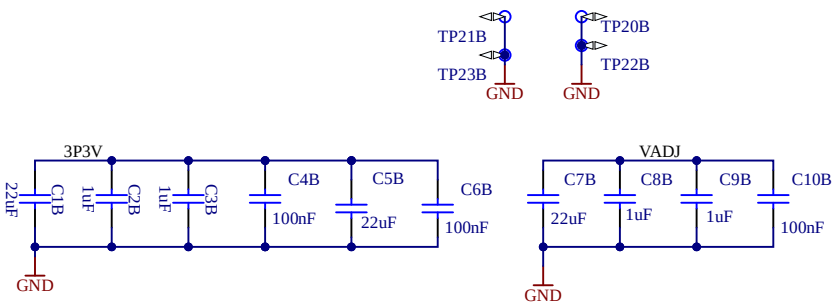
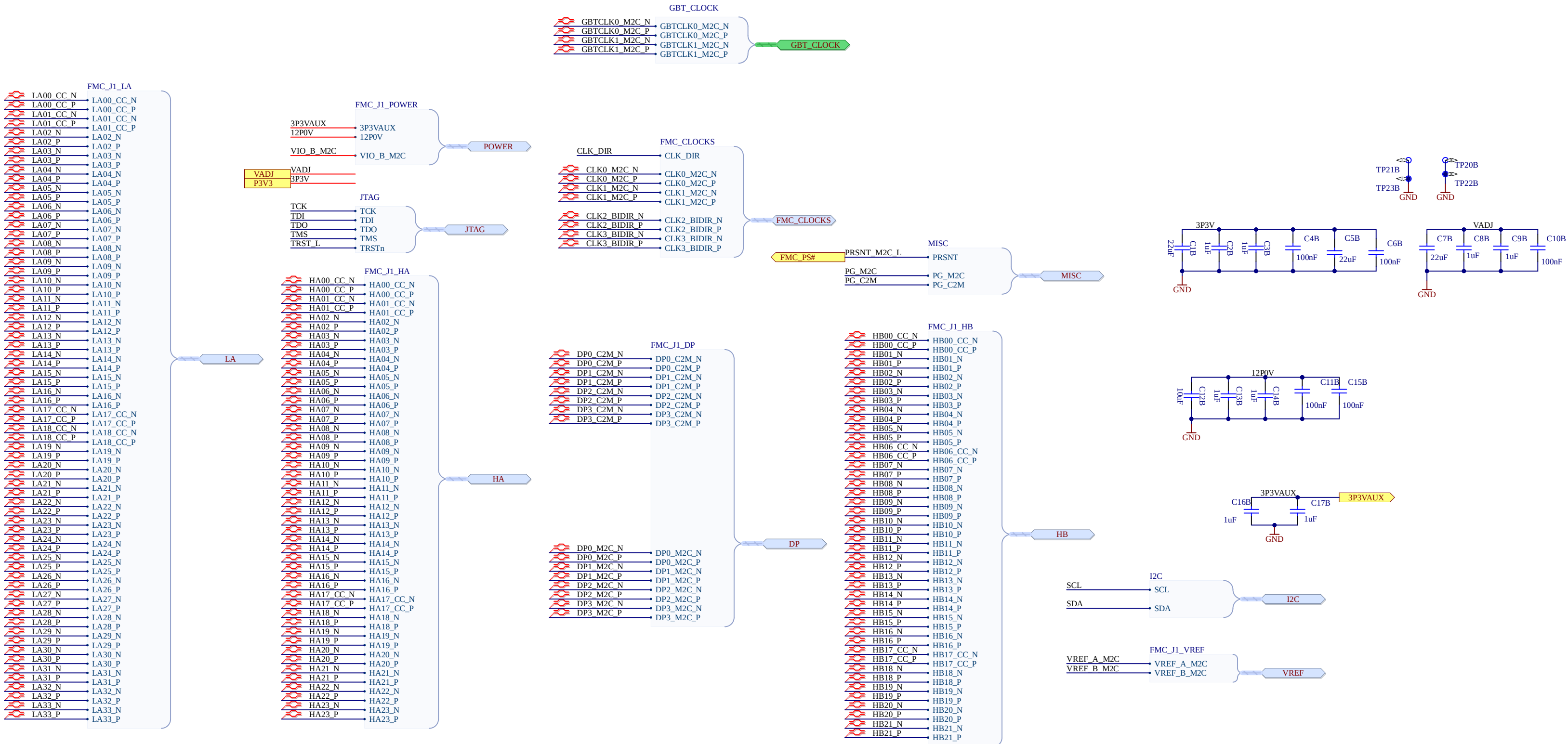
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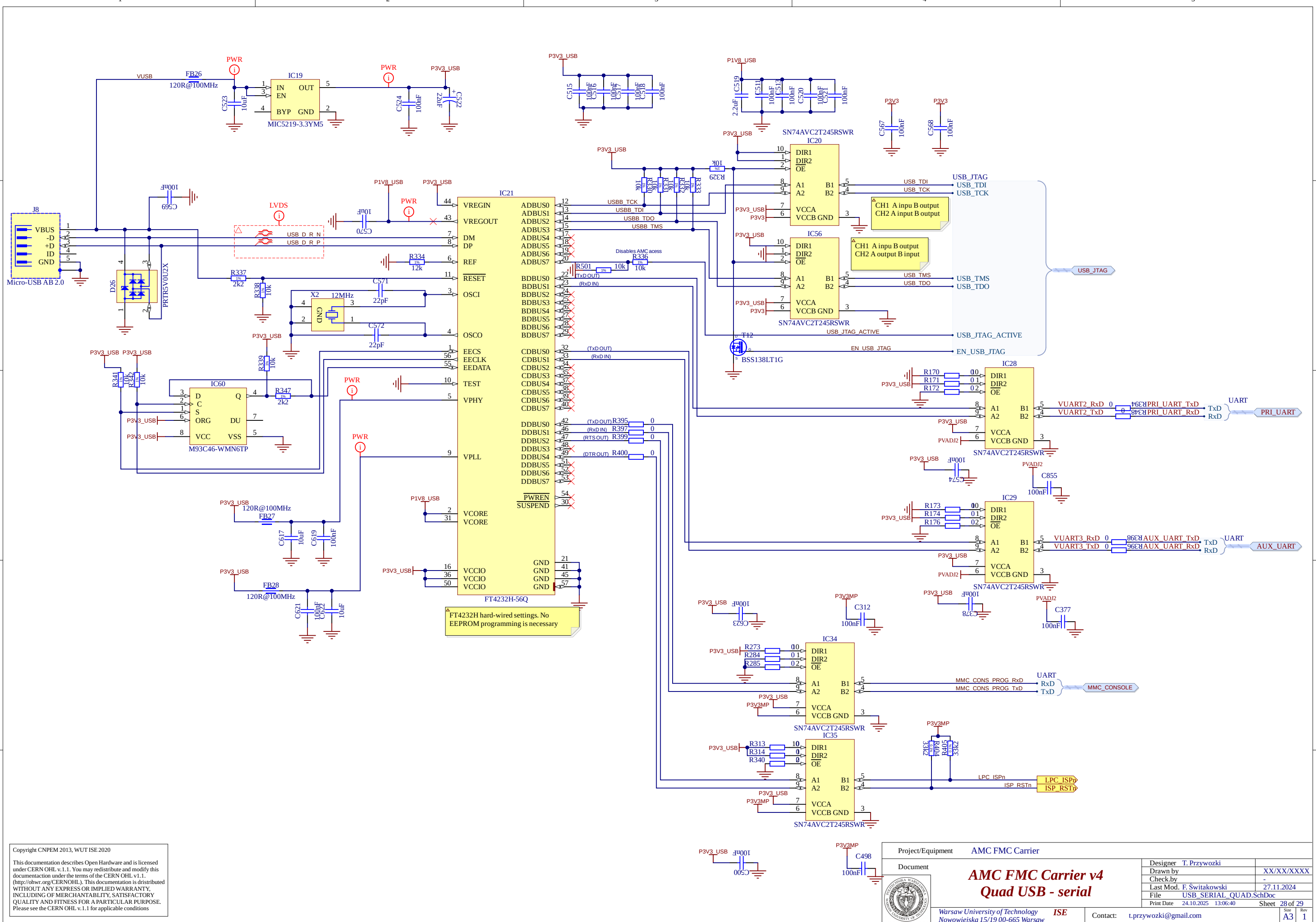
Project/Equipment		AMC FMC Carrier						
	Document AMC FMC Carrier v4 RTCE & EEPROM		Designer	T. Przywozki				
			Drawn by	XX/XX/XXXX				
			Check by	-				
			Last Mod.	F. Świątowski 14.11.2024				
			File	RTCE.SchDoc				
			Print Date	24.10.2025 13:06:39				
	Sheet		26 of 29					
	Warsaw University of Technology Nowowiejska 15/19 00-665 Warsaw		ISE	Contact:	t.przywozki@gmail.com	Size	A4	Rev



VAdj is not defined as power supply and therefore as global signal as it might differ for the 2 FMC



VAdj is not defined as power supply and therefore as global signal as it might differ for the 2 FMC



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A

A

B

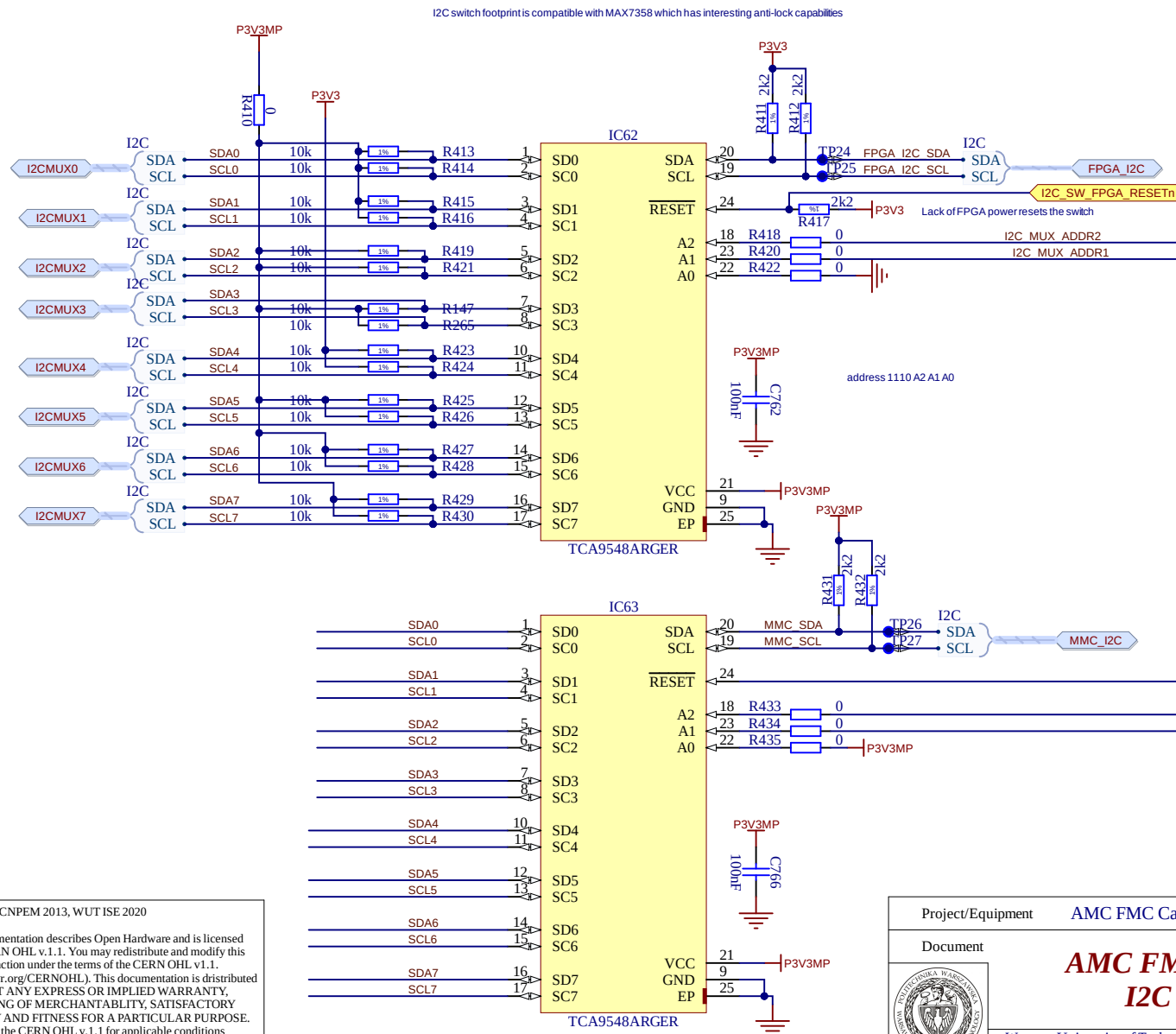
B

C

C

D

D



I2C device address map	MUX Port0	MAX6642 0x48
	Temp sensors	LM75: 0x4C
		LM75: 0x4D
		LM75: 0x4E
		LM75: 0x4F
	MUX Port1	RTC EEPROM: 0x57
	RTCE	SRAM/RTCC reg: 0x6F
MUX Port2		EEPROM 2k: 0x50
		EUI-ID: 0x59
MUX Port3		EEPROM 64k: 0x51
MUX Port4	Power	INA3221AIRGV: 0x40
		INA3221AIRGV: 0x41
		INA3221AIRGV: 0x42
MUX Port5	Clock switch	8V54816ANLG: 0x5B
MUX Port6	RTM	RTM devices
MUX Port7		EEPROM 0x52
		EEPROM 0x50
		FMC1

mux address 1110 A2 A1 0 from FPGA side 0x7x
mux address 1110 A2 A1 1 from CPU side 0x7x

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Project/Equipment AMC FMC Carrier

Document



AMC FMC Carrier v4 I2C switches

Warsaw University of Technology
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ISE

Designer	T. Przywozki	
Drawn by		XX/XX/XXXX
Check by		-
Last Mod.	F. Świtkowski	14.11.2024
File	I2C_MUX.SchDoc	
Print Date	24.10.2025 13:06:40	Sheet 29 of 29

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Size A4 Rev 1